Fast response MPPT charger for the Técnico Solar Boat

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Introduction to the Research in

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I declare that this document is an original work of my own authorship and that it fulfils all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Abstract

Solar-powered vehicles rely on solar energy to charge their batteries and maximize their range, particularly solar-powered boats need an effective energy conversion in rapidly changing environmental conditions. In this context, the Maximum Power Point Tracker (MPPT) charger plays a critical role in energy conversion, by ensuring that the solar panels operate on their Maximum Power Point (MPP) while charging a battery. This work focus on the study, design, simulation, development and test of a fast response MPPT charger for Técnico Solar Boat (TSB).

This thesis present the design and implementation of a MPPT charger with fast response capable of being used in São Gabriel 01 (SG01). The prototype was designed to charge a lithium-ion battery of 43.2 V nominal voltage, converting the energy provided by the solar panel with high efficiency. The proposed charger is based on a synchronous boost converter operating at 100 kHz. An STM32 Microcontroller Unit (MCU) was used to implement the control firmware, including the MPPT algorithm, battery charging state machine, protections, and CAN and serial communication.

Different MPPT techniques and DC-DC converter topologies were analyzed to identify a suitable solution for the dynamic operating conditions of the vessel. Particular attention was given to the trade-off between tracking speed and accuracy, leading to the implementation of a Perturb and Observe (P&O) algorithm. The converter and Photovoltaic (PV) system were simulated in both LTspice and Matlab to validate the component selection and algorithm parameters. A custom Printed Circuit Board (PCB) was designed, manufactured and solder by hand.

The experimental results demonstrate success of the proposed architecture and show that the implemented MPPT algorithm can respond rapidly to changes in irradiance, with the measured tracking response being suitable for the environmental variations identified for SG01. The devolved prototype also provides a practical method to configure the device using a computer application which can also be used for data analysis. The resulting system provides a dedicated, configurable, and low-cost MPPT charger for TSB, while establishing a platform for further improvements in efficiency, tracking performance, and integration into future vessels.

Keywords: MPPT, MPPT topology, DC-DC Converter, Solar energy, P&O, Solar Boat

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Acronyms

ASIC	Application-Specific Integrated Circuit
BMS	Battery Management System
CAN	Controller Area Network
CC	Constant Current
CCM	Continuos Conduction Mode
CV	Constant Voltage
DC	Direct Current
DCM	Discontinuous Conduction Mode
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
FPGA	Field Programmable Gate Array
GUI	Graphical User Interface
IC	Integrated Circuit
IncCond	Incremental Conductance
MCU	Microcontroller Unit
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracker
LUT	Look-Up Table
OC	Over Current

OTS	Off the shelf
P&O	Perturb and Observe
PCB	Printed Circuit Board
PV	Photovoltaic
R&D	Research and Development
SG01	São Gabriel 01
SR03	São Rafael 03
STC	Standard Testing Conditions
TSB	Técnico Solar Boat

1

Introduction

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1.1 Motivation

With environmental pollution intensifying worldwide, the transition to cleaner energy sources such as solar power has become increasingly urgent. In 1985, only 20.82% of the energy produced in the world came from renewable energies. These numbers have been rising every year, and in 2024, have reached 31.92% [11]. In 2024, 6.9% of the energy produced in the world comes from solar panels, and in Portugal this number rises to 14.5%, demonstrating the importance and the impact of solar energy [11]. Solar energy still plays a minuscule role, and it is listed behind the other sources of energy in terms of its contribution to meeting the world's energy demand. However, solar energy is becoming more relevant, with the cost of solar panels dropping significantly in the last decade.

In comparison to other forms of green energy, Photovoltaic energy is relevant due to its availability, simplicity, lower maintenance, environmental friendliness, reliability, and many other benefits. More recently, it is becoming more relevant in the automotive industry, with solar powered cars, boats, and robots [12]. The CO2 emissions of the automotive sector are one of the main contributors to global warming. In 2018, 29% of total CO2 emissions in the EU came from the transport sector, with 4% coming from the maritime sector alone [13]. These challenges have directly motivated the development of the Técnico Solar Boat (TSB) project.

In 2015, TSB was created with the goal of designing and building solar powered boats to compete in international competitions while also contributing to the innovation in yath industry. Since then, the project has grown, and several vessels have been built. It began with the construction of the first solar prototype, São Rafael 01, which had a lot of room for improvement and was followed by São Rafael 02 and 03. The team then decided to approach the hydrogen energy and autonomous driving, and therefore São Miguel 01 and 02 and São Pedro 01 were built. More recently, a vessel was built to combine all the technologies used before and was named São Gabriel 01 (SG01).

All these prototypes used solar energy to maximize their range and efficiency. In the first years the energy produced was small, and the whole system was built using commercial Off the shelf (OTS) components. However since the beginning of the TSB project, the spirit of innovation and the desired to develop all the systems in house has been present. In previous years, the team has built its own batteries, telemetry systems, flight controllers, fuel cell controllers, etc. Some other more complex systems were design during master thesis. Until today a Battery Management System (BMS), dual motor system, Marine Propulsion System, a Maximum Power Point Tracker (MPPT) and more.

In 2020, the team started to build its own solar panels for São Rafael 03. To increase performance and the amount of information acquired by the solar panels' operation, in the same year, surge the first attempt to do a MPPT. This was the fist step to achieve a fully functional product. The result was a working MPPT that was never implemented on any vessel due to constantly burning components which was not reliable. Additionally, the project is outdated in terms of components and firmware, lacking configurability.

The MPPT charger is a fundamental subsystem in photovoltaic energy systems, responsible for maximizing the power extracted from solar panels by ensuring continuous operation at the Maximum Power Point (MPP). A MPPT charger is typically a power converter that continuously monitors the voltage and current produced by the photovoltaic modules and adjusts their electrical operating point accordingly. By dynamically regulating these operating parameters, the MPPT compensates for variations in environmental conditions such as solar irradiance and temperature, which cause the MPP to shift over time. As a result, the photovoltaic system can operate with improved efficiency and deliver maximum available power under changing conditions [14].

In the context TSB project, a solar system is specially important to increase the autonomy of our vessels. In 2023/24 season we were able to consumption of São Rafael 03 (SR03) from 1.5 kW to 300 w by using solar energy Decreasing 80 % of his consumption. With a costume made MPPT, we believe that we could reach even better performances wile also increasing the data acquired about the system performance that would serve to improve the system.

Now in 2025/26 season, the relevancy of the solar system decreased with our new vessel SG01, since it uses around 20 KW of constant power. So the main power source will be hydrogen. However, solar energy will continue to be used in this vessel to power the auxiliary systems being a reliable green source that will keep the boat powered in all situations.

1.2 Objectives

This project aims to design and implement a MPPT charger for the solar panels used in TSB project. The MPPT charger will convert the energy produced by the solar panel as efficiently as possible with the use of a quality DC-DC converter and the implementation of MPP tracking algorithms.

The main objectives of this project are:

- Study and understand the operation of solar panels and MPPT techniques;
- Choose the most suitable DC-DC topology for the system;
- Implement a fast response and accurate control algorithm;
- Design and simulate the hardware and software;
- Implement the system in hardware and develop a Printed Circuit Board (PCB);
- Test and validate the performance of the designed circuit;
- Ensure the safety and reliability of the MPPT charger for its integration in the TSB project;
- Provide performance data of the solar panel and the MPPT through a Controller Area Network (CAN) communication interface;

- Allow easily change the configuration for ease of use.

By achieving these objectives, the project will contribute to a more efficient solar system, maximizing the energy received from the solar panels and increasing the available data for performance tracking of the MPPT and solar panel. This data will be used to later improve the energy efficiency of the MPPT and take conclusions about the manufacture quality of the solar panels build by TSB project. A cheap, efficient and versatile MPPT will also mean future savings to the team, since it could be used in a variety of vessels.

In the context of TSB, such complex projects with high scientific rigor, that involve simulations, PCB design, laboratory tests and field tests are only possible in master thesis. So, for TSB, the main objective is to increase the team knowledge about this subject by increasing the documentation about it. Also, due to a drastic event of another team, which burn all their MPPTs during competition, the security of this system is the main concern.

1.3 Outline

This document is organized into five main chapters:

- **Chapter 1** introduces the context of the work, presenting the motivation behind the development of an MPPT charger for the TSB project. The objectives of the project are defined, and the scope of the proposed solution is established;
- **Chapter 2** presents the State-of-the-Art related to photovoltaic energy systems and MPPT technology. It reviews the operating principles of solar panels, solar energy production, and the most relevant MPPT algorithms. Additionally, different DC-DC converter topologies, battery charging techniques, processing units, and commercial solutions are analyzed and compared;
- **Chapter 3** describes the proposed solution for the MPPT system. The overall system architecture is presented, followed by a summary of the technical requirements. The adopted design approach and methodology are detailed, including converter selection, control strategy, simulation, and experimental validation plan;
- **Chapter 4** presents the preliminary work developed so far, including initial simulations and early design results that support the feasibility of the proposed solution;
- **Chapter 5** outlines the planning and scheduling of the project, presenting the development timeline and key milestones required to achieve the defined objectives.

  Needs to be rewritten

2

System

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2.1 Overview

São Gabriel 01 (SG01) is the most recent vessel built by TSB, which was designed to compete in Sea Lab category of Monaco energy boat challenge. To compete in this category a vessel with 8 x 2.5 m was designed creating new challenges for the team. This vessel was designed to use both hydrogen and solar energy, using hydro foils to hydroplane which highly increases efficiency. Additionally, it will use actuators and sensors to drive autonomously to compete in AI class of Monaco Energy Boat Challenge.

Since it is a larger and heavier boat than the previous ones built by TSB it needs more power than the others. So a motor with 130 KW peak power was chosen. This much power is only possible to provide with high voltages, so for the first time in TSB history this vessel have a High Voltage system (300-600 V) and a Low Voltage system (30-50 V).

The High Voltage system is responsible for the drive train, powering a motor. Since the energy consumed by this motor is much larger than solar panels can ever achieve, the boat is designed to be powered by hydrogen, however it is not implemented yet due to budget constraints.

As for the low voltage system, it is designed to power all auxiliary system (dashboard, telemetry, flight control, electrical string, throttle, etc.). Some of this auxiliary systems are critical for the normal use of the vessel, for example if the dashboard turns off, the pilot will not know the state of the high voltage system (temperature, voltage, state of charge, etc.). Or even more critical, if the throttle does not work, the boat can not electrically move.

The auxiliary systems are always changing and growing, so it is difficult to estimate the power consumption of this system. For this work it is considered 1000 W of continuous consumption. It is not a perfect estimation because the power consumption will depend on the driving mode (autonomous driving, hydroplaning or normal drive) which can create different power profiles due to the actuators. Additionally, the cooling system (which is planed to be moved to the High Voltage system) have high power consumption that is not considered in this estimation.

To help to maintain the low voltage system always powered, it is used solar panels which are connected to several Maximum Power Point Trackers (MPPTs).

For better understanding of the system architecture see Figure 2.1 which represent highly simplified diagram of the overall system.

2.2 PV system

The photovoltaic system of SG01 was designed to maximize the energy harvested from the available deck area while maintaining reliable operation under the varying environmental conditions encountered in marine applications.

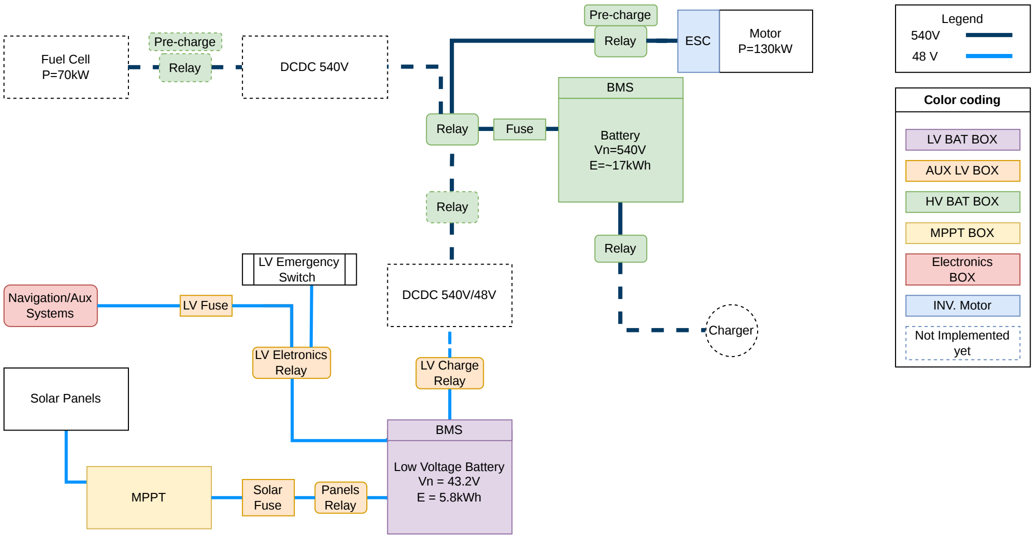
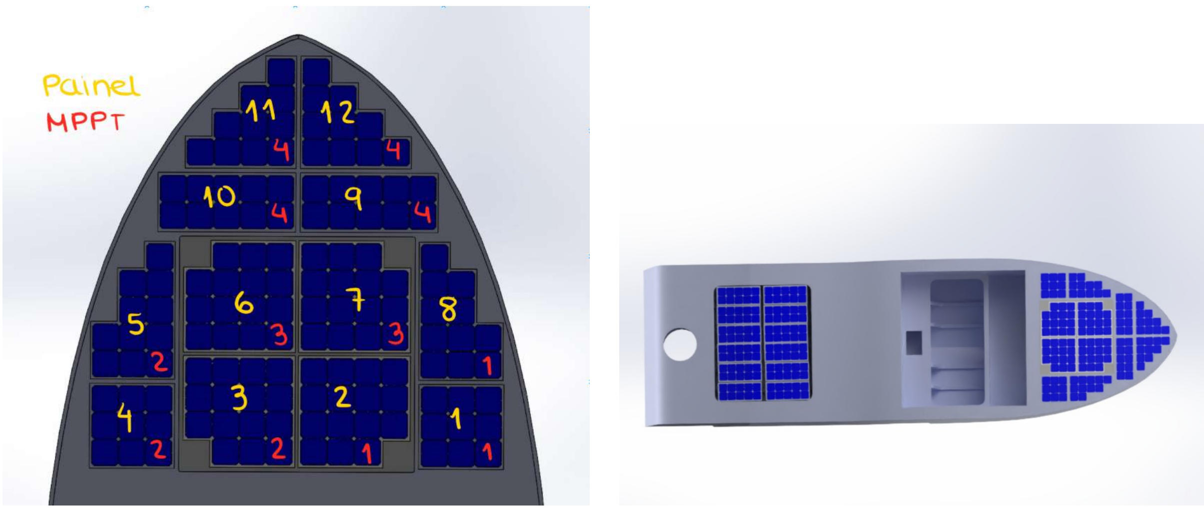


Figure 2.1: Simplified electrical schematic of SG01.

The current installation consists of 140 Maxeon Gen 3 (Me3) photovoltaic cells distributed among ten custom-manufactured solar panels. These panels are connected to four independent MPPTs, as illustrated in Figure 2.2a. The distribution of panels among the MPPTs was carefully selected to reduce the effects of partial shading, temperature gradients, and non-uniform solar irradiance across the vessel surface. By electrically separating regions with different operating conditions, each MPPT can independently track the optimal operating point of its associated panels, increasing the overall energy yield.



(a) Distribution of Solar Panels and MPPT on SG01.

(b) Render of São Gabriel 01 with additional solar panels.

Figure 2.2: Solar panels installation on SG01.

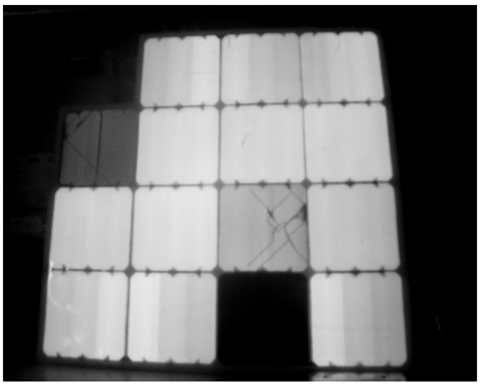
Each photovoltaic cell has an active area of approximately 155.06 cm^2, resulting in a total photovoltaic area of 2.171 m^2.

Assuming Standard Testing Conditions (STC) with an irradiance of 1000 W/m^2, the photovoltaic array is capable of delivering approximately 526.4 W of peak power (theoretical value).

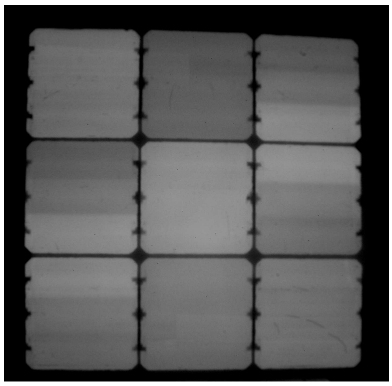
The photovoltaic energy is processed by four commercial Genasun MPPTs, which are grouped in a dedicated MPPT box. The MPPTs convert the variable voltage produced by the solar panels into a stable charging voltage suitable for the low-voltage battery system. This architecture allows each photovoltaic group to operate close to its maximum power point independently, improving energy extraction under non-uniform illumination conditions that are common in maritime environments.

To ensure high performance and shape configurability, the photovoltaic modules were manufactured in house using materials selected specifically for marine operation. This process begins by soldering all cells in a module together and then a lamination is performed using vacuum and heat blacks.

Prior to installation, all panels underwent electroluminescence testing to detect manufacturing defects such as microcracks, broken cells, or interconnection failures, Figure 2.3. These tests were performed before integration into the vessel to ensure that only panels with acceptable electrical and structural integrity were installed. This quality-control procedure reduces the likelihood of premature degradation and enables comparison before and after competitions.



(a) Worst results.



(b) Best results.

Figure 2.3: Electroluminescence test results.

Although the current photovoltaic installation already provides a significant contribution to the low-voltage energy budget, additional usable deck area remains available. Figure 2.2b presents a conceptual rendering with supplementary solar panels. Preliminary estimates indicate that the total photovoltaic power could be increased to approximately 977 W, almost doubling the currently installed capacity. Such an upgrade would substantially reduce battery charging times and significantly increase the operational autonomy of the vessel.

2.3 Battery

In the past years we have been producing our batteries achieving a lower cost and highly customizable solution. In SG01 we chose lithium-ion cells due to its ease of use quick assembly. The assembly is done by the members. It begins by placing the cells on plastic holder to increase structural strength, then using a spot welder, the series and parallel connections are made with nickel tabs (Figure 2.4). In parallel, some thermistors are place in different locations of the battery to ensuring a precise a robust temperature reading. In the end, the battery is placed in a case with a Battery Management System (BMS) and all the necessary connections.

Figure 2.4: Low voltage battery assembly.

The current low voltage battery uses Samsung INR21700-50G cells in a 12s28p configuration. This configuration results in the specifications written in Table 2.1.

SG01 battery was designed for a different vessel which needed a substantial higher capacity. The battery is too heavy, and overall overdimensioned.

For lab tests and for a more realistic impact analysis, SR03 battery will be used. This is a much smaller battery with a reasonable capacity for this vessel. It uses Samsung INR21700-50s cells in a 12s7p configuration. This configuration results in the specifications also written in Table 2.1.

2.4 PV System Impact on Autonomy

With SG01 battery (5844.56 Wh) and the maximum power that the solar panels can currently provide (526.4 W), the minimum time to fully charge the battery is 11h and 7min (5844.56 Wh / 526.4 W). This

Table 2.1: SG01 and São Rafael 03 (SR03) battery specs.

Parameter	Unit	SG01 Battery	SR03 Battery
Max Voltage	V	50.4	50.4
Nominal Voltage	V	43.2	43.2
Min Voltage	V	30	30
Configuration	–	12s28p	12s7p
Number of cells	–	336	84
Energy Capacity	Wh	5855.56	1512
Capacity	Ah	140	35
Max charge current	A	135.8	42
Cutoff charge current	A	3.395	1.75
Max discharge current	A	271.6	175
Weight	kg	25.2	?
Internal resistance	mΩ	6	24

is a large time to charge a battery. There are two solutions to this problem: increase the power provided by the solar panels or decrease the battery capacity.

Let's consider the case where we reduced the battery capacity of 1512 Wh which is the capacity of our solar boat (SR03) battery. With this change alone the battery is fully charged in 2h and 53min, which is a much more reasonable time. Additionally, if we increase the power provided by the solar panels to 977 W, the battery will be fully charged in 1h and 33min, which is an excellent time to charge a battery.

The charging times are a good to evaluate the Photovoltaic (PV) system, but the main goal is to maximize the time on water before full battery discharge. For that lets consider 1000 W of consumption. Considering the original battery it takes 5 h 51 min to fully discharge without solar panels and 12 h 21min with solar panels and approximately 10 days with the additional solar panels. As previously said, this battery is overdimensioned and will be changed in the future.

To get a better idea of the impact of the PV system lets consider the battery of SR03. This battery takes 1.51 h to fully discharge without solar panels and 3 h 12 min with solar panels and approximately 2 days 18 h with the additional solar panels.

By reducing the battery capacity and increase the solar panel area, the battery would never need external charging. In another point of view, the low voltage system can drain more power from the battery.

2.5 Environmental changes

In engineering, the term "fast", used in the title of this thesis, needs to be associated with a time scale. A process can only be considered fast when compared with another process with a longer characteristic time. Therefore, this section investigates the characteristic time scales associated with variations in the Maximum Power Point (MPP) caused by environmental changes, with particular emphasis on the oscillatory motion of the vessel and its implications for the required speed of the MPPT algorithm.

When analyzing MPP variations, there two main factors: temperature and irradiance. Temperature variations are generally slower than variations in irradiance. Although rapid temperature changes can occur, for example when water comes into contact with the photovoltaic panels, such events are considered unusual during normal operation and are therefore not considered in this analysis. Irradiation on the other hand, can vary more rapidly as the angle of the solar panels changed with the vessel angle, particularly roll and pitch.

The motion of a vessel subjected to waves can be described as a forced dynamic system. The vessel has characteristic natural frequencies associated with its different modes of motion, including roll, pitch and heave. The response to wave excitation depends on the relationship between the excitation frequency and the corresponding natural frequency of the vessel. In particular, when the excitation frequency approaches a natural frequency, the response amplitude can increase significantly due to resonance. The vessel does not, however, necessarily oscillate at its natural frequency when subjected to excitation at another frequency; the steady-state response occurs primarily at the excitation frequency, while the natural frequency determines the dynamic response and its amplitude.

Considering a simplified undamped free oscillation system, the natural frequency of a vessel can be calculated using the natural period Formula 2.1 where I is the mass moment of inertia and C is the restoring stiffness.

$$T = 2\pi\sqrt{\frac{I}{C}} \tag{2.1}$$

When applied to pitch and roll, the restoring stiffness is given by the mass (m), gravity (g) and metacentric height (GM_L for pitch and GM_t for roll). Since the inertia mass was not given, the equation was further simplified using the radius of gyration in pitch or roll (k_{yy} or k_{xx}) which is equivalent to Formula 2.2.

$$k = \sqrt{\frac{I}{m}} \tag{2.2}$$

$$T = 2\pi\sqrt{\frac{I}{mgGM}} \leftrightarrow T = 2\pi\sqrt{\frac{k_{yy}^2}{gGM_L}} \tag{2.3}$$

For pitch, $k = k_{yy}$ and $GM = GM_L$, whereas for roll, $k = k_{xx}$ and $GM = GM_T$.

This formula is used with results given by the hydrostatic simulations which were provided by Técnico Solar Boat (TSB), where GM_L is 21.515 unit and GM_t is 1.867 unit. Also, the radius of gyration is calculated based on the longitude (L=8 m) and width/beam (B=2.5 m). Based on the International Towing Tank Conference (ITTC) recommendation [15], when the radius of gyration for pitch and roll is unknown a value of 0.25L and 0.35B to 0.40B, respectively, should be used.

Based on the given values, the pitch and roll natural period is 0.8654 and 1.2853 respectively, which is equivalent to 1.1555 Hz and 0.7780 Hz. These values represent estimates of the natural frequencies of the vessel.

As for the frequency of the waves it has an extensive range and depends on several factors. Studies on this subject show higher energy for waves with 1 to 30 seconds periods which is equivalent to 0.033 Hz to 1 Hz, Figure 2.5.

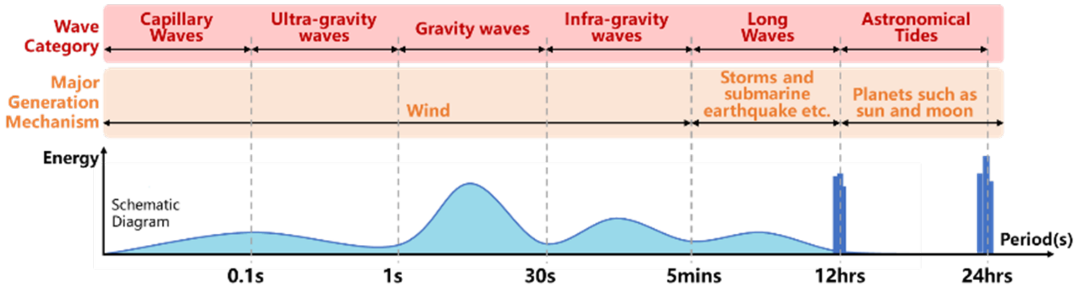


Figure 2.5: Classification of waves by period and respective energy [1].

The environmental excitation experienced by SG01 was further investigated using experimental data collected while the vessel was adrift at sea. Ten datasets containing the roll and pitch angles were collected and combined to obtain the frequency-domain response shown in Figure 2.6. The results show significant spectral components at approximately 0.05 Hz and 0.5 Hz. the lower frequency can be explained by the fast winds, and it is consistent with the local forecast wave period. The faster frequency can be caused by the natural frequency of vessel which in real conditions can be lower from what was calculated.

Both theoretical and experimental results predict or measure frequency up to 1 Hz. In the context of the MPPT algorithm, the goal will be a minimum speed of 2 Hz or 0.5 seconds of tracking time to give some margin.

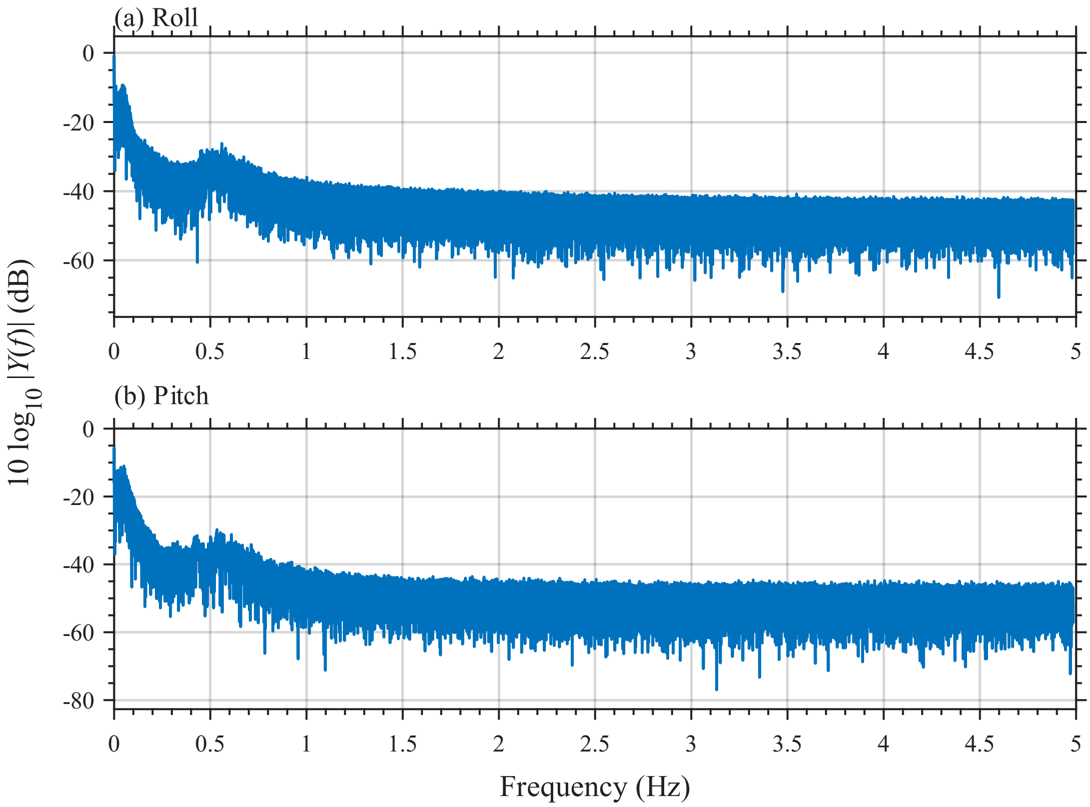


Figure 2.6: Measured oscillations from SG01.

3

State-of-the-Art

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In this chapter, the main State-of-the-Art concepts regarding the engineering behind an MPPT charger and solar panels are explained. More specifically, it addresses key challenges such as: what is a solar panel, MPPT algorithms, DC-DC converter types, and battery charging techniques.

3.1 Solar Panels

Photovoltaic (PV) panels are devices that convert solar irradiance into electrical energy. A PV panel consists of multiple solar cells electrically connected in series and/or parallel to achieve the desired voltage and current levels. As in conventional electrical systems, the interconnection topology directly affects the electrical characteristics of the panel: series connections increase the output voltage, while parallel connections increase the output current [16].

Series-connected PV panels operate at higher voltage levels, which reduces conduction losses in interconnecting cables and improves the efficiency of the DC-DC conversion stage. Moreover, power converters designed for high voltage and low current operation are generally less complex and more cost-effective than those operating at low voltage and high current. However, series configurations are more sensitive to partial shading or panel failure, as the output current of the entire string is limited by the weakest module [17]. In contrast, parallel configurations provide improved tolerance to shading and faults but result in higher current levels and increased conduction losses. In this project, series/parallel configuration of the solar panels will not be changed, since it is already pre-defined by Técnico Solar Boat (TSB), but it is important to understand the implications of each configuration.

Each cell is made of semiconductor materials, usually silicon. This semiconductor is doped with phosphorus, a group V element, to create a negative type layer. On the other side, a layer is doped with boron, a group III element, to create a positive type layer. This creates a p-n junction, which is essential for the photovoltaic effect [3].

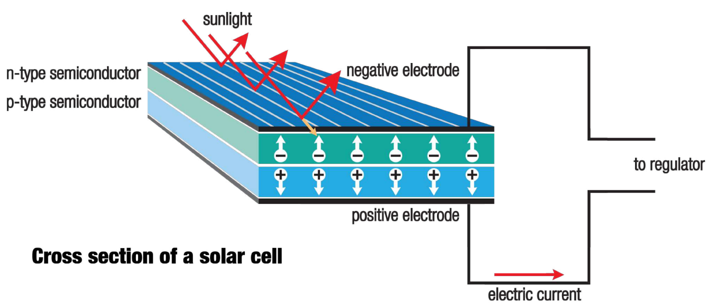


Figure 3.1: Principle of operation of a solar panel [2].

When sunlight arrives at the solar cell, the photons energy is absorbed by the semiconductor material. This energy excites electrons, allowing them to escape their atomic bonds and create electron hole pairs. The electric field at the p-n junction drives these free electrons towards the n-type layer and holes towards the p-type layer, generating a flow of electric current when the cell is connected to an external circuit [3], Figure 3.1.

The energy generated results in voltage and current outputs that are not constant. Voltage and current are interdependent, therefore, a variation in one parameter leads to a corresponding variation in the other. This relationship is nonlinear and can be characterized by a current-voltage (I-V) curve. Furthermore, the power output also varies with operating conditions and can be represented by a power-voltage (P-V) curve. Both curves are represented in Figure 3.2. In both curves, there is a specific point where the power output is maximized, known as the Maximum Power Point (MPP). This point corresponds to a unique combination of voltage and current, denoted as V_{mpp} and I_{mpp}, respectively. Operating the solar panel at this point ensures optimal energy extraction under a given condition.

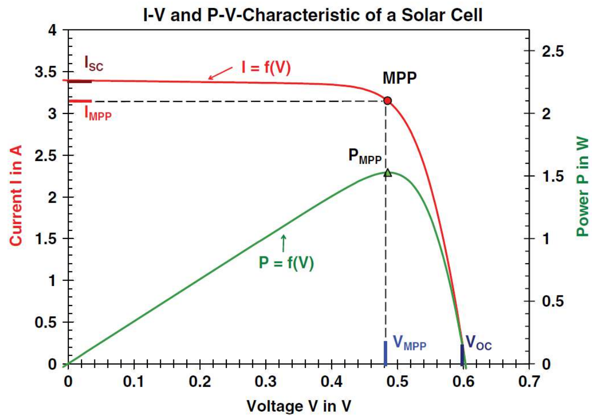
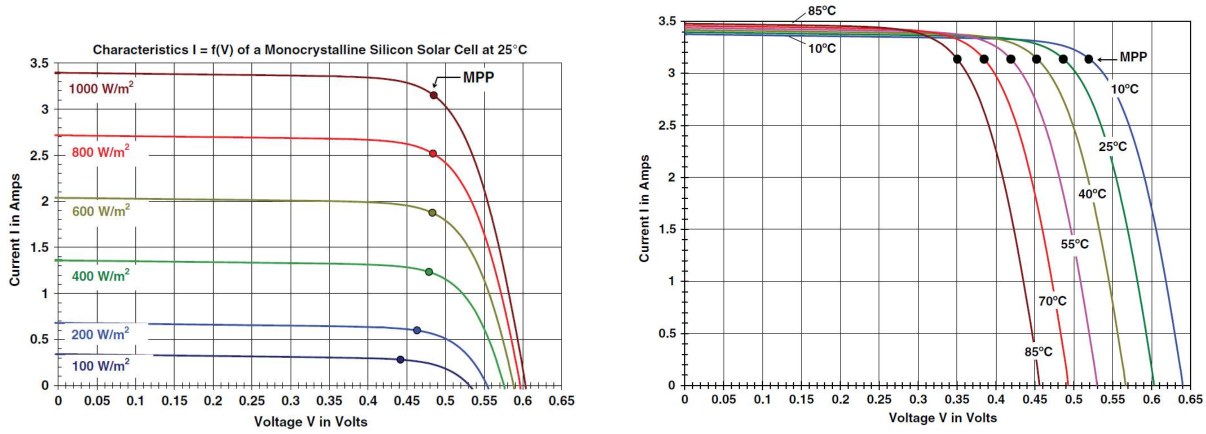


Figure 3.2: I-V and P-V curves of a solar panel [3].

The characteristic curves of a solar panel depend on various factors, including the quality of the materials used, the solar cells, and environmental conditions such as temperature, solar irradiance, wind, and shades. It is noted that under partial shading conditions, it is possible to have multiple local maxima, but overall, there is still only one true MPP [18].

As seen in Figure 3.3a, the variation of the solar irradiance changes the I-V and P-V curves of the solar panel. The increase in irradiance produces more power by mainly increasing the current output of the panel. On the other hand, the temperature has an opposite effect (Figure 3.3b). The increase in temperature produces a decrease in power output, mainly by decreasing the voltage output of the panel.

Both environmental conditions have a meaningful impact on the MPP of the solar panel and therefore on the power output of the system. To maximize the power extracted by the solar panels, a classic power converter, like a buck converter with fixed duty cycle, will not be efficient since neither the current nor the voltage are constant for different conditions. Instead, an MPPT charger is used, which will find the MPP and operate the solar panel at that point while sensing the changes caused by the environment.



(a) Cell temperature of 25 ºC and variable Irradiance.

(b) Irradiance of 1 $\mathrm{kW/m^2}$ and variable temperature.

Figure 3.3: I-V curves under different conditions [3].

3.2 Maximum Power Point Tracker (MPPT)

Refer to moving MPPTs/ mecanical MPPT

A Maximum Power Point Tracker (MPPT) charger is a device that is used to optimize the power output from solar panels by continuously tracking and adjusting the operation point of the panels to ensure they operate at their MPP.

To achieve its goal, an MPPT charger is usually composed of a DC-DC converter, a microcontroller, and some sensors (Figure 3.4). The sensors are used to measure the input/output variables of the control system (typically voltage and current). The information acquired by these sensors is processed by the microcontroller, which runs an algorithm to determine if the MPP was reached or how to act on the system towards the MPP. In the second case, the microcontroller generates a signal to control the DC-DC converter that will adjust its operation accordingly.

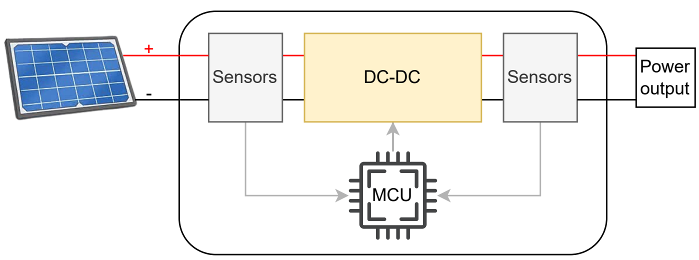


Figure 3.4: Simplified structure of an MPPT charger.

The output can be connected to different types of loads, like batteries, DC loads or inverters. This project focuses on batter charging applications.

3.3 MPPT Algorithms

There are plenty of MPPT algorithms, each one with its own advantages and disadvantages. The choice of the algorithm will depend on the specific application, the desired performance, and the available resources. In this work, the aim is to extract the maximum energy possible of a solar panel installed in a moving boat, which will produce an I-V curve variation due to the quick changes in irradiance caused by changes of inclination, orientation and clouds, and also temperature changes due to waves and wind. For these reasons, tracking speed and accuracy are the most important factors to consider.

With this in mind, the most relevant algorithms are briefly reviewed in order to identify the most suitable option for this project.

3.3.1 Constant Voltage (CV)

This is the simplest and most inefficient method. It works by operating the solar panel at a fixed voltage. This referenced voltage is used to calculate the duty cycle of the DC-DC converter that will maintain the operating point of the solar panel.

Well-known variants improve performance by setting the reference voltage to a fraction of the open-circuit voltage (typically 71-78%) [18]. This voltage is obtained by briefly disconnecting the panel from the load, increasing reference accuracy and efficiency. Another variant uses a fraction of the short-circuit current as the reference [19].

Although both methods are simple and cost-effective, they are inherently inefficient due to the necessity of interrupting energy production during measurement phases. However, alternative implementations address this limitation through proxy measurement techniques employing dummy cells or diodes, whose physical properties closely approximate those of standard solar cells, thereby enabling continuous power generation during measurement acquisition [18] [20]. Still, the MPP is not always located at the same percentage of the open-circuit voltage or short circuit current, leading to a lack of accuracy and therefore lower efficiency [14].

3.3.2 Perturb and Observe (P&O)

The Perturb and Observe (P&O) method is widely used in commercial products and is the basis of many advanced algorithms. Its popularity lies in its simplicity, low cost, and ease of implementation.

As its name indicates, the algorithm perturbs the voltage of a PV array and observes the resulting effect on the output power [21]. If an increase in voltage leads to an increase in output power, the operating point is moving toward the MPP, and the algorithm continues to increase the voltage. Conversely, if the output power decreases, the operating point is moving away from the MPP, and the direction of the perturbation is reversed [4, 22]. Figure 3.5 illustrates the flowchart of the algorithm.

The biggest drawback of this algorithm is that it oscillates around the MPP, which produces power losses. This oscillation can be reduced by decreasing the step size of the perturbation, but this will also reduce the tracking speed of the algorithm. This results in an inherent trade-off between tracking speed and accuracy [18, 19].

To mitigate this issue, an adaptive step size can be used, where the step size is larger when the MPP is far away and smaller when it is closer. Consequently, the tracking speed is maximized while minimizing the oscillations around the MPP. This can be achieved by measuring the power and voltage and calculating the P-V curve slope ($\Delta P/\Delta V$). At the MPP, this slope approaches zero, allowing the step size to be reduced accordingly. This adaptive variant is commonly referred to as the differential power perturb-and-observe(dP-P&O)".

Another weal-know issue of P&O is that it can get confused in rapidly changing environmental conditions, like fast irradiance changes caused by clouds. In this case, the algorithm can misinterpret the power change caused by the environmental variation as a result of its own perturbation, leading it to move away from the MPP instead of towards it [18]. For example, if the perturbation was in the wrong direction, but the irradiance increased, the power output would increase, and the algorithm would continue perturbing in the wrong direction (Figure 3.6).

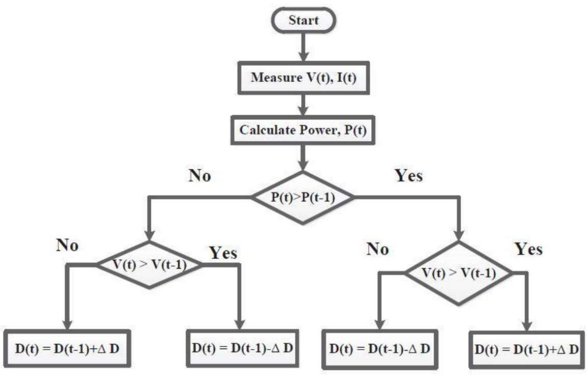


Figure 3.5: Flow chart of the classic version of Perturb and Observe algorithm [4].

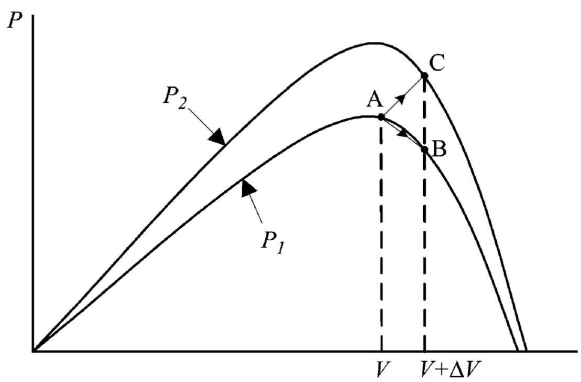


Figure 3.6: Divergence of P&O from MPP as shown in [5].

To address this limitation, an additional condition is incorporated into the algorithm that evaluates two consecutive measurements of ΔP and ΔV. Specifically, when the signs of consecutive ΔP measurements do not follow the signs of ΔV, this indicates that the observed power variation results from environmental disturbances rather than the algorithm's perturbation. Consequently, the voltage adjustment is suppressed. This refined variant is designated the two-point algorithm or improved P&O method [16].

There are other variations of P&O algorithm that improve its performance, like Variable Step Size P&O (VSS-P&O), the three-point P&O, a-factor P&O, and more. Unfortunately, due to the scope of this work, it is not possible to explain all of them.

3.3.3 Incremental Conductance (IncCond)

The Incremental Conductance (IncCond) algorithm is another widely used MPPT method due to its accuracy and ability to track the MPP under rapidly changing environmental conditions.

This algorithm is based on the fact that at the MPP, the derivative of power with respect to voltage is zero. By knowing the output voltage and current of the solar panel, the algorithm can calculate the conductance and the incremental conductance [23].

The following formulas, Equation 3.1, show the equation on which this algorithm is based.

$$\frac{dP}{dV} = \frac{d(V \cdot I)}{dV} = I + V\frac{dI}{dV} \rightarrow \frac{1}{V} \times \frac{dP}{dV} = \frac{I}{V} + \frac{dI}{dV} \tag{3.1}$$

At the MPP point, the slope of the P-V curve is zero, which means that the negative of the conductance, $-I/V$, is equal to the incremental conductance, dI/dV, (Figure 3.8). The algorithm compares these two values to determine if the operating point is at, to the left, or to the right of the MPP [19] [18]:

$$\begin{aligned}
\frac{dP}{dV} = 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} = -\frac{I}{V} \qquad \text{at MPP} \\
\frac{dP}{dV} > 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} > -\frac{I}{V} \qquad \text{left of MPP} \\
\frac{dP}{dV} < 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} < -\frac{I}{V} \qquad \text{right of MPP}
\end{aligned} \tag{3.2}$$

Based on these conditions, the algorithm adjust its operating point (increasing or decreasing the voltage) using a variable step size based on the instantaneous conductance relative to the incremental conductance, Figure 3.7. For example, as suggested in [18], the step size can be calculated using a proportional integral (PI) controller with zero as input reference and error defined as:

$$e = \frac{I}{V} + \frac{dI}{dV} \tag{3.3}$$

Comparative analysis demonstrates that the IncCond algorithm exhibits superior performance relative to the P&O method. This outcome is expected, as the Incremental Conductance technique was developed specifically to address the limitations inherent in the P&O approach [23]. For example, the IncCond method eliminates the oscillation problem around the MPP that is characteristic of the P&O method. Additionally, IncCond demonstrates superior performance under rapidly changing environmental conditions, such as sudden irradiance variations, since it uses derivative-based information that is less sensitive to transient disturbances [16].

Note: Figure 3.7 was made for a buck converter, so the duty cycle actions are inverted for a boost converter. For example, if the operating point is to the left of the MPP, the duty cycle should be decreased to increase the voltage of the solar panel and move towards the MPP.

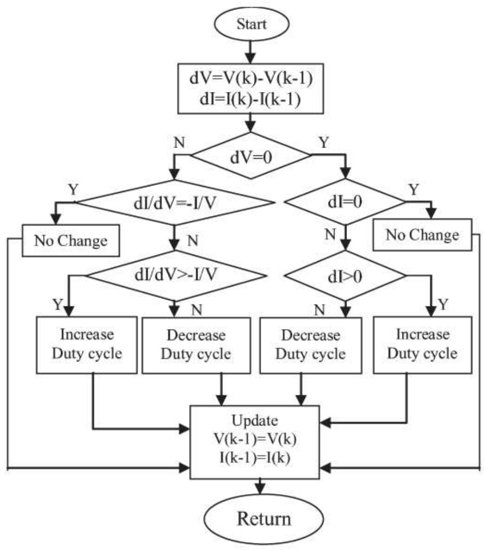


Figure 3.7: Flowchart of the IncCond method with direct control [6].

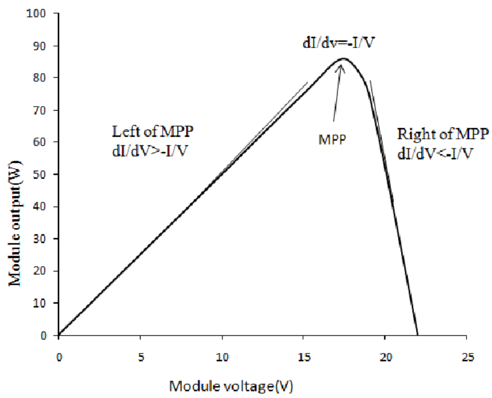


Figure 3.8: IncCond method principle.

3.3.4 Look Up Table (LUT)

The Look-Up Table (LUT) method is a simple and fast MPPT algorithm that relies on pre-calculated/measured data to determine the optimal operating point of a solar panel [19].

The LUT table contains several entries organized by voltage and current. Each of these entries contains the optimal duty cycle of the DC-DC converter, that are pre-calculated or measured for a specific voltage and current. This approach allows the algorithm to reach the MPP in one clock cycle by measuring the voltage and current of the solar panel, looking for the closest entry in the LUT table, and applying the corresponding duty cycle to the converter. The entries of the table can be calculated through a mathematical model of the solar panel with different conditions (temperature and irradiance) or measured in a laboratory.

Table 3.1 is a visual representation of a 2D LUT, only based on current and voltage. Each entry represents the duty cycle used in each case.

The main advantage of this method is its speed since it can reach the MPP in one clock cycle. This makes it suitable for applications where fast tracking is required, like in this project. However, it is not very accurate [4].

To improve the accuracy, the LUT table needs to be larger, which increases the required memory resources. Also, by adding the temperature and/or irradiance to the table would increase the accuracy, making it almost 100% accurate, but this would increase the size of the table exponentially. In extreme cases, the memory requirements became expensive and slow. There is a trade-off between size, speed, and accuracy that needs to be considered when using this method. Also, the LUT method does not adapt

Table 3.1: LUT matrix representation with $m \times n$ elements.

	V_1	$\cdots$	V_k	$\cdots$	V_m
I_1	D_{11}	$\cdots$	D_{1k}	$\cdots$	D_{1m}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_j	D_{j1}	$\cdots$	D_{jk}	$\cdots$	D_{jm}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_n	D_{n1}	$\cdots$	D_{nk}	$\cdots$	D_{nm}

to the aging of the solar panel or changes in its characteristics over time, unless the table is updated periodically through new measurements.

3.4 Types of DC-DC Converters

The DC-DC converter is a crucial component of an MPPT charger, as it allows the adjustment of the operating point of the solar panel to match the MPP. His choice is highly important to the project since it directly affects the efficiency, cost, and complexity of the overall system.

DC-DC converters can be classified into two main categories: isolated and non-isolated converters. The main difference between these two types of converters is the presence or absence of galvanic isolation between the input and output circuits [24].

3.4.1 Non-isolated versus isolated DC-DC converters

This galvanic isolation is usually achieved by using a transformer, which provides electrical separation between the input and output sides of the converter. This isolation is important in applications where safety is a concern, such as in medical devices or industrial equipment, as it helps to prevent electrical shock and damage to sensitive components. In the context of solar energy systems and E-mobility, isolated converters are often used when the solar panel is connected to the grid [25], as they help to protect against voltage spikes and other electrical disturbances. Also, non-isolated converter can provide very higher/lower gain, which is useful for applications where solar panels MPP voltage is very different from the battery or load voltage (typically medium/high voltages which is not the case).

In contrast, non-isolated converters do not provide galvanic isolation between the input and output circuits. However, they generally offer higher efficiency, lower cost, reduced size, and lower circuit complexity (Table 3.2). Additionally, they present fewer challenges in thermal management and are simpler to design. These advantages, together with the fact that galvanic isolation is not a requirement for this application, led to the exclusion of isolated DC-DC converters from this project.

Table 3.2: Comparison between Non-isolated and isolated converters.

Type	Non-isolated Converters	Galvanic Isolated Converters
Circuit Complexity	Low	High
Efficiency	High	Medium
Size	Compact	Large
Electromagnetic Interference (EMI)	Medium/High	High
Switching Frequency	Medium	Medium
Thermal Management	Easier	Moderate
Control Complexity	Moderate	Moderate
Gain	Near 1	Very High/low
Cost	Low	High

Legend: undesired medium good

3.4.2 Non-isolated DC-DC converters

Modern non-isolated conversion circuits generally use one of three basic topologies: buck, boost, or buck-boost converters. They are "basic" in the sense that only one switching element is needed. A given topology is used to obtain a specific result, such as voltage step-down, voltage step-up, or hybrid mode [8].

The **boost converter** is a step-up DC-DC converter widely employed in MPPT controllers when the required output voltage is higher than the solar panel open circuit voltage. It uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is on the OFF state, the inductor releases the stored energy to the output through the diode, increasing the output voltage (Figure 3.9b) [26].

This topology has been extensively documented in the literature and is distinguished by its inherent simplicity, cost-effectiveness, and superior efficiency characteristics [12] [4] [23]. Depending on the design variation used and constraints, it can achieve conversion efficiencies up to 98% [27].

A limitation of this topology is that it cannot emulate an impedance lower than the load impedance, which prevents it from reaching values close to the short-circuit current of the PV module. Consequently, it is not compatible with all solar panels or MPPT algorithms [27, 28].

The **buck converter** is a step-down DC-DC converter also widely used for applications where the solar panel voltage is higher than the battery or load voltage. Similar to the boost converter, it uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. However, in this case, the input is connected to the output when the transistor is in the ON state. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, decreasing the output voltage (Figure 3.9a) [27].

This converter also has a similar downside to the boost converter. The buck converter can not emulate a smaller impedance than the load impedance, and therefore, it can not reach values near the open circuit voltage of the PV module [27] [28].

Buck-Boost converters are step-up and step-down DC-DC converters that can operate in both modes. This makes them more versatile than the previous two topologies, but also more complex and less efficient [27]. This converter uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, increasing or decreasing the output voltage (Figure 3.9c) [8].

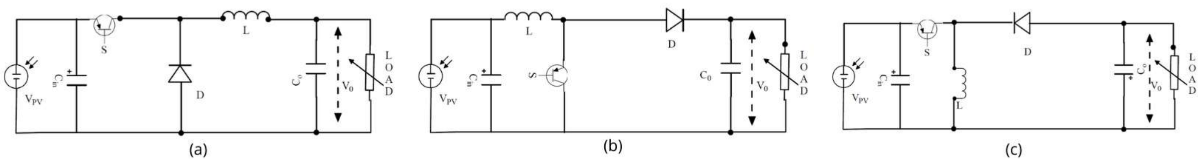


Figure 3.9: Buck (a), Boost (b) and Buck-Boost (c) DC-DC converters.

The conventional buck-boost converter produces an output voltage inverted with respect to the input which creates challenges on sensor measurements, driving circuit and isolation, consequently, it is unsuitable for the battery charging system [26].

An enhanced variant of the buck-boost converter is the CUK converter, illustrated in Figure 3.10a. The main characteristic of the CUK converter is that the currents through inductors L_1 and L_2 are equal, allowing the use of a common magnetic core, which helps reduce current ripple. However, a major drawback of this topology is that the entire load current must pass through the coupling capacitor C. In addition, the converter produces an inverted output voltage [26].

The issue of output voltage polarity inversion can be addressed by using a single-ended primary inductor converter (SEPIC), as shown in Figure 3.10b. This makes the SEPIC converter suitable for battery charging applications. Nevertheless, its main drawbacks are the high output current and voltage ripple which increases losses and electromagnetic interference (EMI) [29]. These limitations are mitigated by the ZETA converter, shown in Figure 3.10c, which provides improved output current characteristics while maintaining the same output voltage polarity as the SEPIC converter [22] [26].

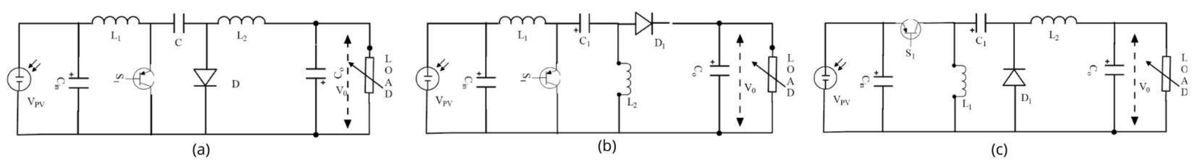


Figure 3.10: CUK (a), SEPIC (b) and ZETA (c) DC-DC converters.

3.4.3 Comparison between Non-isolated DC-DC converters

All the non-isolated DC-DC topologies explained before have their own advantages and disadvantages. The choice of the topology will depend on the specific application, the desired performance, and the available resources. To help with this choice, Table 3.3 shows a comparison between the most relevant characteristics of each topology [29]. The most significant advantages of each topology are highlighted accordingly with the aim of the project.

Table 3.3: Comparison of non-isolated DC-DC topologies.

Topology	Boost	Buck	Buck-Boost	ZETA	CUK	SEPIC
Electrical performance						
Efficiency	High	High	High	Medium/High	Medium	Medium
EMI	Low	Low	High	Medium	Medium	Medium
Input current ripple	Low	High	High	Low	Very low	Medium
Output current ripple	High	Low	High	Very low	Low	Medium
Output voltage ripple	Medium	Low	Medium	Low	Medium	High
Structural characteristics						
Circuit complexity	Low	Low	Low	Medium	Medium	Medium
Size	Low	Low	Low	Medium	Medium	Medium
Storage elements	1 L	1 L	1 L	2 L, 1 C	2 L, 1 C	2 L, 1 C
Switch stress	No	Yes	No	Yes	Yes	Yes
Functional aspects						
Type	Step-Up	Step-Down	Step-Down/Up	Step-Down/Up	Step-Down/Up	Step-Down/Up
Versatility	Low	Low	Medium	High	Medium	High
Inverted output	No	No	Yes	No	Yes	No
Reliability	High	High	High	Medium	Medium	Medium
Economic						
Cost	Low	Low	Low	Medium	Medium	Medium

Legend: good important

For MPPT controllers, the choice is mainly governed by the possibility of operating at the MPP which is connected with the converter type (Step Up, Down or Down/Up). Also, efficiency, cost (related to the number of storage elements), output voltage ripple are relevant in this choice.

The buck converter can be excluded from this comparison since in this project the solar panel voltage is lower than the battery voltage, therefore, a step-down converter is not suitable. However, when choosing between the boost and the step-up/down converters, there is a trade-off. The boost converter is simpler, cheaper and more efficient. On the other hand the step-down/up converters are more versatile since they can operate in a wide range of input voltages, which is interesting since the size of the solar panels and battery voltage can change.

Considering that in this project the solar panel voltage is always lower than the battery voltage, the boost converter is the most suitable topology due to its simplicity, cost-effectiveness, and high efficiency. However, if versatility were to become a priority, the ZETA converter would be the preferred choice among the step-up/down topologies, owing to its non-inverted output voltage and low output current ripple, which are advantageous for battery charging applications.

3.5 Battery charging techniques

As the demand for electronic devices and E-vehicles increases, the need for efficient, compact, and lightweight batteries has emerged. Among many existing technologies, lithium-ion batteries have one of the best energy-to-weight/volume ratios and, at this moment, it is the technology used in TSB batteries. But, like every other battery technology, they need to be charged properly to ensure their safety and longevity [7].

To charge these batteries properly, the MPPT algorithm can operate alone if an additional converter is used in series. But in most of MPPT chargers, the MPPT and the battery charging unit are integrated in the same converter. This way, the MPPT algorithm can adjust the operating point of the solar panel to extract the maximum power while the battery charging technique ensures that the battery is charged properly. A proper combination of both algorithms/techniques is essential to ensure both maximum energy extraction and battery safety.

There are several techniques to charge lithium-ion batteries, the most relevant for this use case being the constant current-constant voltage (CC-CV) method [30]. Also, it is widely used in commercial chargers due to its simplicity and effectiveness [31]. This method consists of two main phases (Figure 3.11). In the first phase, the battery is charged with a constant current until it reaches a predefined voltage limit (usually 4.2 V per cell). In the second phase, the voltage is held constant at this limit while the current gradually decreases as the battery approaches full charge. Once the current drops below a certain threshold, the charging process is terminated to prevent overcharging [31].

When employing the CC-CV charging technique with a MPPT controller, the initial Constant Current (CC) phase is regulated by the MPPT algorithm, which extracts the maximum available power from the solar panel limited by the maximum charging current. Once the Constant Voltage (CV) phase is reached, the MPPT function is disabled, and the power converter increases the solar panel voltage to progressively reduce the charging current while keeping the output voltage constant.

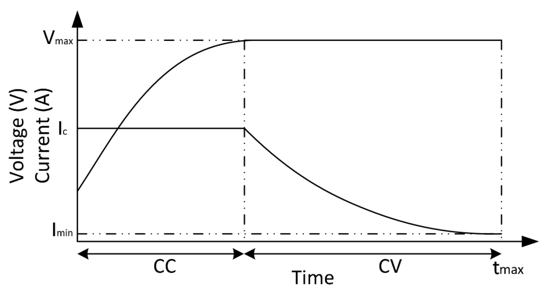


Figure 3.11: Charging profile of CC/CV [7].

More advanced high-performance battery charging techniques have been proposed in literature to reduce charging time without increasing the charging current, thereby preserving battery lifetime. However, these techniques generally introduce higher switching losses and increased system complexity. In the present application, the current generated by the photovoltaic panels is less than half of the maximum allowable battery charging current, consequently, the charging current is inherently limited by the solar panels rather than by the charging strategy. Furthermore, battery lifetime is not a critical concern in this project, as the battery is expected to undergo a limited number of charge cycles, not exceeding 20 cycles per year.

Therefore, the CC-CV charging method is considered the most suitable technique for this project, owing to its simplicity, effectiveness, and widespread adoption in commercial battery chargers.

Other charging techniques fall outside the scope of this work, however, some methods are briefly discussed in the references provided [30] [7] [31].

3.6 Commercial Options

To assess the feasibility of adopting a Off the shelf (OTS) solution, this section reviews existing commercial options for solar battery charging with MPPT capability. Both specialized Integrated Circuits (ICs) and complete commercial MPPT chargers are considered, and their electrical, functional, and economic characteristics are evaluated against the requirements of this project. The objective is to identify whether a commercially available solution can satisfy the constraints of the application or, alternatively, to justify the need for a custom-designed converter and control system.

3.6.1 Specialized ICs

Several commercially available Integrated Circuit (IC) are specifically designed for solar battery charging and MPPT applications. These devices are relevant to this project since, if suitable, they could significantly reduce design complexity by integrating functions such as DC-DC conversion, protection mechanisms, and MPPT algorithm.

Table 3.4 summarizes four commonly used commercial ICs. Characteristics marked in red indicate incompatibilities with the project requirements, leading to the exclusion of the corresponding devices.

The BQ24650RVAT is a Stand-Alone Synchronous Buck Battery Charge Controller which implements a high-efficiency battery charger while also providing safety features. By adding a Microcontroller Unit (MCU) to run a MPPT algorithm and current and voltage sensors, this IC can track the MPP of a solar panel, making it appropriate for applications with 12/24 V batteries and high-power panels, such as electric vehicles. However, it does not support the required 48 V battery voltage and is therefore unsuitable for this work.

The MAX20801TPBA+ and SPV1040TTR target low-power IoT applications with small batteries and solar panels. Both devices are unable to meet the 48 V output requirement, and their limited publicly available information on tracking speed further restricts performance evaluation.

The LT8491IUKJ#PBF offers a wide input and output voltage range, buck-boost operation, high efficiency, communication capabilities, and CC-CV charging. Despite these advantages, its reported MPPT tracking speed of up to 2 s is insufficient for the dynamic conditions of this application. Additionally, its I^2C interface would require an additional microcontroller to provide CAN communication, increasing system cost.

Table 3.4: Comparison of commercial specialized ICs.

Model	BQ24650RVAT	MAX20801TPBA+	SPV1040TTR	LT8491IUKJ#PBF
Electrical specifications				
Input voltage	5 to 28 V	1.5 to 18 V	0.3 to 5.5 V	6 to 80 V
Output voltage	12 to 24 V	12.4 V	-0.3 to 5.5 V	1.3 to 80 V
Max output current	Externally limited	12 A	1.8 A	10 A
Switching frequency	600 kHz	Unknown	100 kHz	100 to 400 kHz
Electrical efficiency	95%	99.1% (max)	80 to 95%	95 to 99%
MPPT and control				
Algorithm	User input	Unknown	Unknown	P&O
Tracking accuracy	User dependent	99.9% (max)	Unknown	Unknown
Tracking speed	User dependent	Unknown	Unknown	1 to 2 s
Functional aspects				
Topology	Synchronous Buck	Synchronous Buck	Synchronous Boost	Buck-Boost
Communication	No	No	No	I^2C
Configurability	No	No	No	Yes
Charge profile	CC-CV	Unknown	Unknown	CC-CV
Extra hardware needed				
Transistors	Yes	No	No	Yes
Sensors	Yes	No	No	No
Storage elements	Yes	Yes	Yes	Yes
Economic				
Price	5.25 €	4.50 €	3.02 €	15-20 €

Legend: unsuitable

3.6.2 Commercial MPPT chargers

Several commercial MPPT chargers are available that can address the problem under analysis without requiring additional hardware. However, each solution presents limitations when evaluated against the project requirements. Table 3.5 summarizes selected commercial MPPT chargers, with red-marked characteristics indicating incompatibilities.

The GVB-8-Li-CV(50.4) is a high-performance and fast MPPT charger, particularly suited for dynamic environments such as marine applications. It has been successfully used by TSB in previous projects, demonstrating high efficiency and reliability. Nevertheless, its lack of configurability, fixed output voltage, and high cost restrict its suitability for this application. Despite these drawbacks, it serves as a valuable reference design.

The Smart Solar MPPT 100/20, developed by Victron Energy, offers high efficiency, fast tracking, and integrated communication interfaces. However, it uses a buck topology that requires higher input voltages, making it incompatible with the small and distributed solar panel arrays used in the TSB application.

Other commercial solutions, such as the Rover Lite and MS4840N, provide acceptable performance but rely on communication protocols that are not compatible with the project requirements and offer limited technical documentation.

The Reboost V0.2.1 is an open source MPPT charger originally developed for automotive applications. While it provides flexibility and near adequate specifications, its reliance on expensive and discontinued components, along with the use of a boost topology, limits scalability and adaptability to potential future battery voltage changes.

Overall, none of the evaluated commercial MPPT chargers fully satisfy the technical, economic, and configurability requirements of this project, motivating the development of a custom solution.

Table 3.5: Comparison of commercial MPPT controllers.

Model	**GVB-8-Li-CV (50.4V)**	**Smart Solar MPPT 100/20**	**Reboost V0.2.1**	**Rover Lite**	**MS4840N**
General information					
Brand	Gensun	Victron Energy	TPEE	Renogy	BougeRV
Type	Boost	Buck	Synchronous Boost	Boost	Boost
Control and communication					
Communication	No	VE.can and Bluetooth	Yes	Bluetooth and RS485	Bluetooth
Configurable	No	Yes	Yes	Yes	Yes
GUI	No	Yes	Yes	Yes	Yes
MPPT and charging					
Prog. battery voltage	No	Yes	Yes	non-lithium batteries	Yes
Tracking speed	15 Hz / 66.6(6) ms	Fast (unknown)	Programmable	Unknown	Unknown
Tracking efficiency	99%+ typical	Unknown	Programmable	99%	Unknown
Electrical efficiency	96-99% typical	98% peak	Unknown	97%	Unknown
Charger profile	CC-CV	Unknown	Programmable	Unknown	CC-CV
Hardware implementation					
MCU	ATtiny461A-U	Unknown	STM32G474	Unknown	Unknown
MCU Details	8-bit, RISC, 64 MHz	–	32-bit, Arm, 170 MHz	–	–
Transistor	FZT951	Unknown	GS61008T	Unknown	Unknown
Transistor type	PNP BJT	Unknown	GaN FETs	Unknown	Unknown
Economic					
Price	240 €	100-200 €	Components: 250-280 € PCB: 20-100 € Total: 270-380 €	300-350 €	120-160 €

Legend: unsuitable

4

Implementation

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introdução do capítulo

4.1 DC-DC converter

introdução do capítulo

4.1.1 Operation principles

The DC-DC converter chosen for this project was a boost converter. This topology uses one switching device, one inductor, one output capacitor and a diode, Figure 4.1. The switching device is used to control the output voltage by switching on and off at a specified frequency (fsw) and duty cycle (D). The inductor is used to store energy when the switch is on (stage 1) and release it when the switch is off (stage 2), while the output capacitor is used to smooth the output voltage, Figure 4.1.

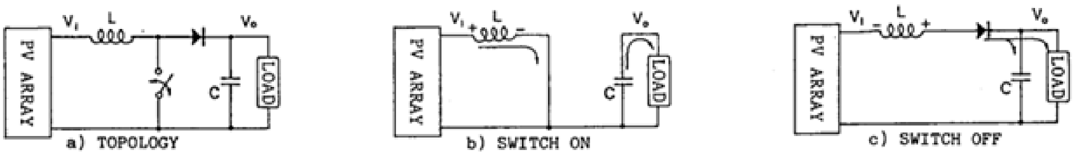

Figure 4.1: Classic Boost converter topology and stages [8].

Analyzing the boost converter separately on his two stages, it is possible to define the drop voltage across the inductor (VL), as a function of the input voltage (Vin) and output voltage (Vout) as follows:

$$V_L = \begin{cases} V_{in} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.1}$$

With equation 4.1 and the fact that at stationary state the average voltage across the inductor is zero, it is possible to define the output voltage in function of the input voltage and duty cycle as follows:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} V_{in} dt + \int_{DT}^{T} (V_{in} - V_{out})dt) = 0 \Rightarrow V_{out} = \frac{V_{in}}{1-D} \tag{4.2}$$

Equation 4.2 is one of the most important equations in boost converter design, as it defines the relationship between the input voltage, output voltage and duty cycle. As the duty cycle increases, the output voltage also increases, which is the main principle behind boost converters. For our use case, it is import to analyze the relation between the input and output current as well. Considering no power losses, it is possible to define the input current as follows:

$$P_{in} = P_{out} \Leftrightarrow V_{in} \cdot I_{in} = V_{out} \cdot I_{out} \Leftrightarrow I_{in} = \frac{V_{out}}{V_{in}} \cdot I_{out} \Leftrightarrow I_{in} = \frac{I_{out}}{1-D} \tag{4.3}$$

However, these equations are only valid for ideal components since it does not take into account the parasitic elements of the components, such as the resistance of the inductor, the drain source resistance of the transistor, and the voltage drop across the diode (there are more parasitic components that affect the performance). In a real boost converter, these parasitic elements will cause a voltage drop and therefore reduce the output voltage.

Let's quickly analyze the influence of the inductor resistance (R_L) and the voltage drop across the diode (V_d) by changing the equation 4.1 to:

$$V_L = \begin{cases} V_{in} - R_{DS_{(on)}} \cdot I_{DS} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} - V_d & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.4}$$

In this case the at stationary state the average voltage across the inductor is $R_L \cdot I_L$ and therefore the output voltage can be defined as:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} (V_{in} - R_{DS_{(on)}} \cdot I_{DS})dt + \int_{DT}^{T} (V_{in} - V_{out} - V_d)dt) \Leftrightarrow \tag{4.5}$$

$$\Leftrightarrow R_L \cdot I_L = \frac{1}{T}[T \cdot D \cdot (V_{in} - R_{DS_{(on)}} \cdot I_{DS}) + T \cdot (1-D) \cdot (V_{in} - V_{out} - V_d)] \Leftrightarrow V_{out} = \frac{V_{in} - R_{DS_{(on)}} \cdot I_{DS} - R_L \cdot I_L}{1-D} - V_d \tag{4.6}$$

Equation 4.6 shows the 3 mains causes of losses in a boost converter. The main cause is the voltage drop across the diode, which causes a significant reduction in the output voltage (usually 0.3 V to 0.7 V). This will be solved by using a synchronous boost converter (Figure 4.2), which replaces the diode with a transistor. In synchronous boost converters the voltage drop across the transistor is much smaller due to the low $R_{DS,on}$, usually lower than 0.1 V.

As for the inductor internal resistance, it is not possible to remove it entirely, so it must be taken into account when designing the boost converter. The same is true for the drain source resistance of the transistor, which is also not possible to remove entirely. Therefore, it is important to select components with low internal resistance to minimize the losses and increase the efficiency of the converter.

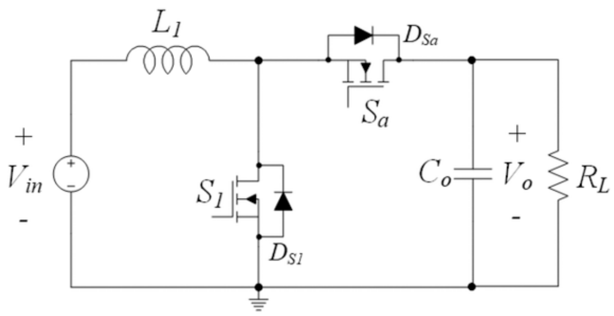

Figure 4.2: Synchronous boost converter topology [9].

4.1.2 Dimensioning

Let us first define the design goals of the synchronous boost converter. The converter must be able to step up the voltage from the solar panels, which may consist of a maximum of 40 cells (Maxeon Gen 3 from SunPower) connected in series, a minimum of 20 cells in series, and is optimized for 35 cells (Figure 4.1). The converter must charge a lithium-ion battery with a nominal voltage of 43.2 V. The switching frequency was defined as 100 kHz. This relatively high frequency allows the use of smaller inductors and capacitors, while still avoiding excessive switching losses.

Table 4.2 and 4.1 summarizes the system requirements.

Table 4.1: PV panel parameters at MPPT.

Configuration	V_{mppt} (V)	I_{mppt} (A)	P_{mppt} (W)
40 cells	24.68	6.05	139.6
35 cells	21.875	6.05	122.15
20 cells	12.34	6.05	69.8

Table 4.2: Converter design constraints.

Parameter	Value
Battery voltage (nominal)	43.2 V
Battery voltage (max)	50.4 V
Battery voltage (min)	30 V
ΔV_o	0.1 V
ΔV_i	0.5 V
$\Delta I_i n$	0.6 A
Efficiency	$>$ 90%

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Since the load is a battery, the output ripple voltage must be limited to avoid aging and heating. The industry standard is to limit the output voltage ripple to 0.5% of the nominal voltage. As for Therefore, the values presented in Table 4.2 were selected. Also, the voltage output voltage ripple must be low to avoid unexpected behavior of the electronic components that are also connected to the battery. As for the input (the solar panel) the voltage ripple must be low for easier measurement and precise control. As for the current, it must be low to reduce electromagnetic interference and losses.

4.1.2.A Inductor

The inductor is one of the most important components in the boost converter since it is responsible for storing energy and releasing it to the load.

The choice of an inductor will highly influence the performance of the converter. High inductance will generate low current ripple, however it generates losses due to its internal resistance. On the other hand a smaller inductance will generate high current ripples, high stress in components and high magnetic interference.

On the other hand, a smaller inductance will generate high current ripples that can cause the system to operate in Discontinuous Conduction Mode (DCM). In DCM the inductor current reaches zero during the switching period, which means that the energy stored in the inductor is not enough to supply the load during the off time. This can cause high current and voltage ripples, which can damage the components and reduce the efficiency of the converter. Therefore, it is important to ensure that the inductance is high enough to operate in Continuos Conduction Mode (CCM).

The equation that describes the inductor is defined as:

$$V_L = L \cdot \frac{dI_L}{dt} = L \cdot \frac{\Delta i_l}{\Delta t} \Leftrightarrow L = \frac{V_L \Delta t}{\Delta i_L} \tag{4.7}$$

In the first stage (transistor in conduction mode), the voltage in the inductor is equal to the input voltage and it occurs for $D \cdot T$ seconds. Therefore, the ripple current in the inductor can be defined as follows:

$$\Delta I_L = \frac{V_{in} \cdot D \cdot T}{L} \tag{4.8}$$

As for the current ripple, it can be described using the riple factor and the average current in the inductor as follows.

$$r = \frac{\Delta I_L}{I_{L,avg}} \tag{4.9}$$

$$I_{L,avg} = \frac{I_o}{1 - D} \tag{4.10}$$

Using Equation 4.10, 4.9, 4.7 and 4.2, the following equation is achieved:

$$L = \frac{V_{in}(1 - D)DT}{I_o \cdot r} = \frac{V_{out}(1 - D)^2 DT}{I_o \cdot r} \tag{4.11}$$

The value of D that maximizes Equation 4.11 is 1/3, which results in the critical inductance formula:

$$MAX[(1 - D)^2 \cdot D] = (1 - 1/3)^2 \cdot 1/3 = \frac{4}{27} \tag{4.12}$$

$$L_{critical} = \frac{4}{27} \cdot \frac{V_o}{I_o \cdot r \cdot f} \tag{4.13}$$

Using the maximum battery voltage (50.4 V) and the maximum current ($I_{pv} + Margin = 10\ A$), and a ripple factor of 2 which is equivalent to the CCM limit we achieve the following result:

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 2 \cdot 100000} = 3.73 \mu H \tag{4.14}$$

This is a reference value, if it were used it would highly stress the components, generating heat, magnetic interference and high current ripples. Therefore, the calculation was repeated aiming the output current ripple of 0.6 A, which is the value defined in Table 4.2. Solving equation 4.9 in terms of the output current ripple, the following equation is achieved:

$$r = \frac{\Delta I_L}{I_{L,avg}} = \frac{\Delta I_{out} \cdot (1-D)}{\frac{I_{out}}{(1-D)}} = \frac{\Delta I_{out} \cdot (1-D)^2}{I_{out}} \tag{4.15}$$

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 0.4 \cdot 100000} = 18.66 \mu H \tag{4.16}$$

The final value of the inductor was defines as 68uH 100 μH, which results in a maximum inductor ripple current of 1.26 A and an output current ripple of 0.53 A. refazer o calculo

To select the apropriate inductor, the rated current and the saturation current must be taken into account. The rated current is the maximum current that the inductor can handle without overheating. In this case the rated current must be higher that the maximum short circuit current of the biggest solar panel configuration (40 cells in series), which is 6.56 A. Some margin must be added to this value, so the rated current must be higher than 10 A.

The saturation current is the maximum current that the inductor can handle without saturating. When the inductor saturates, the inductance drops significantly, which can cause high current ripples and damage the components. Therefore, the saturation current must be:

$$I_{sat} > I_{in} + \frac{\Delta I_L}{2} = 6.56 + \frac{1.26}{2} = 7.19\ A \tag{4.17}$$

The inductor chosen was the 74437529203680 from Würth Elektronik, which has an inductance of 68 μH, a rated current of 16 A and a saturation current of 7.2 A ($|\Delta L/L| < 10\%$) and 10.6 A ($|\Delta L/L| < 30\%$). The saturation current is too close with the objective, however since the inductance is higher than what is needed, the inductance would still be acceptable even at -30% inductance.

Additionally, it has a low Equivalent Series Resistance (ESR) of only 10.3 $m\Omega$, high self resonant frequency of 5.7 MHz and high Operating voltage of 250 V.

4.1.2.B Capacitor

The output capacitor is responsible for smoothing the output voltage and reducing the voltage ripple. The capacitor electric characteristic can be described by the following equation:

$$i_c = C \frac{dv_o}{d_t} \tag{4.18}$$

Based on this equation, the capacitor value can be calculated based on the desired voltage ripple and the load current as follows:

$$i_c = C_o \frac{\Delta V_o}{\Delta t} = C_o \frac{\Delta V_o}{D \cdot T} \Leftrightarrow C_o = \frac{I_o D}{\Delta V_o \cdot f} = \frac{I_{in}(1-D)D}{\Delta V_o \cdot f} \tag{4.19}$$

Using the maximum input current of 10 A, a duty cycle of 0.5 (value that maximizes the formula), a voltage ripple of 0.1 V and a switching frequency of 100 kHz, the following value is achieved:

$$C_o = \frac{10 \cdot (1 - 0.5) \cdot 0.5}{0.1 \cdot 100000} = 250\mu F \tag{4.20}$$

The same can be done for the input capacitor, but in this case the voltage ripple is not as critical as in the output capacitor, since the input voltage is not directly connected to the load. Therefore, a voltage ripple of 0.5 V was used:

$$C_{in} = i_{in} \cdot \frac{\Delta t}{\Delta V_i} = \frac{i_{in} \cdot D}{\Delta V_{in} \cdot f} = \frac{10 \cdot 0.5}{0.5 \cdot 100000} = 100\mu F \tag{4.21}$$

To choose the appropriate capacitors, the voltage rating, current ripple and ESR must be taken into account. The voltage rating must be higher than:

$$V_{rating} > MAX(V_{in}) + \Delta V_{in} = 28.5 + 0.5 = 29\ V \tag{4.22}$$

Where 28.5 is the maximum voltage of a solar panel (40 cells in series) and 0.5 is the voltage ripple defined in Table 4.2.

The current ripple must higher than the RMS input current, which can be calculated based on the RMS formula for triangle wave as follows:

$$I_{ripple} > \frac{\Delta I/2}{\sqrt{3}} = \frac{1.26/2}{\sqrt{3}} = 0.36\ A \tag{4.23}$$

As for the ESR the value must be as low as possible to reduce the power losses.

After analyzing the market, the input capacitor selected was the 870575875006 which is a 56 μF (2 in parallel) with a voltage rating of 63 V and a ripple current of 2.4 A each (total of 4.8 A). The main reason for this choice was his low ESR (22 $m\Omega$) compared to other capacitors with similar values. Additionally, by using two capacitors in parallel, the ESR is reduced even further and the current ripple that they with stand is higher and the reliability of the system is increased.

The same process was repeated for the output capacitor, but in this case the voltage rating must be higher than the maximum battery voltage, which is 50.4 V. The current ripple must be higher than the RMS output current, which can be calculated as follows:

$$V_{rating} > MAX(V_{out}) + \Delta V_{out} = 50.4 + 0.1 = 50.5\ V \tag{4.24}$$

$$I_{ripple} > \sqrt{\frac{I_{in}^2}{4} + \frac{\Delta I^2}{12} \cdot (1 - D)^2} = \sqrt{\frac{6.56^2}{4} + \frac{1.28^2}{12} \cdot (1 - 0.27)^2} = 3.29\ A \tag{4.25}$$

The ripple current in the output capacitor is much higher than in the input capacitor, and it is not a common value for electrolytic capacitor, therefore instead of using a single capacitor, multiple capacitors in parallel will be used to achieve the desired current ripple rating.

After simulations and analyzing the market, the output capacitor value were increased due to the output current ripple dependency on the output voltage ripple. The final capacitor chosen was the 860040875005 with 100 μF (5 in parallel, so 5x100 μF) with a voltage rating of 100V and a ripple current of 0.8 A each (total of 4 A). Some other options were considered, and compared for optimization in section 5.1.2.

Additionally, the output and input needs to filtered to reduce the high frequency noise generated by the switching. This is achieved by adding a small capacitor in parallel with the main capacitors. The values use are 1 μF and 0.1 μF. For this use case ceramic capacitors are better since their stacked multilayer construction provides extremely low ESR and Equivalent Series Inductance (ESL) which over all let them maintain low impedance up into the MHz range.

4.1.2.C Transistor and gate driver

The transistor is the switching device on a synchronous boost converter. The choice of the transistor will highly influence the behavior and performance of the converter.

When selecting a transistor for a switching converter, not only the maximum rating mater but the switching speed and internal resistance are also quite important.

There are many types of transistors (IGBT, MOSFET, GaN, etc.), however, at this frequency (100 kHz) and power level the choice is between MOSFETs and GaN transistors (Figure 4.3). The GaN (Gallium Nitride) transistor technologies has appeared in 1993 in lab experiments, however it has only reached the market more recently, first for radio frequency products in 2004 and only in 2009 for power electronics. The advantage of GaN transistors is their low internal resistance and very fast switching speed, which can significantly reduce the switching losses and therefore increasing the efficiency of the converter. Additionally, they have a high energy density and lower Vgs(on). However, they are more expensive and require a precise gate driver to operate correctly because there max(Vgs) is usually low.

In applications where the efficiency is a key factor in the design like solar inverters or Maximum Power Point Tracker (MPPT), it is better to use the component that has lower Internal (On) Resistance (RdsOn) and supports higher switching frequency. In this case, GaN would be a better fit. MOSFET or Silicon Carbide MOSFET would take the second place since MOSFETs can handle higher switching frequencies as compared to IGBTs [10].

There are a lot of options in the market, however there was one in specifically that standout due to their compact size, while having integrated gate driver and two transistors in the same package. The Integrated Circuit (IC) is the EPC23102, which was designed for power applications, with an incredibly low RdsOn (only 5.2 $m\Omega$), 35 A of maximum power and a maximum operating voltage of 100 V.

Additionally, it has incorporates shoot-through protection that protects the transistor from a control error, however a dead time needs to be introduced in software. It uses a DFN package which is good for heat dissipation and can always be attached to a heat sink for better performance.

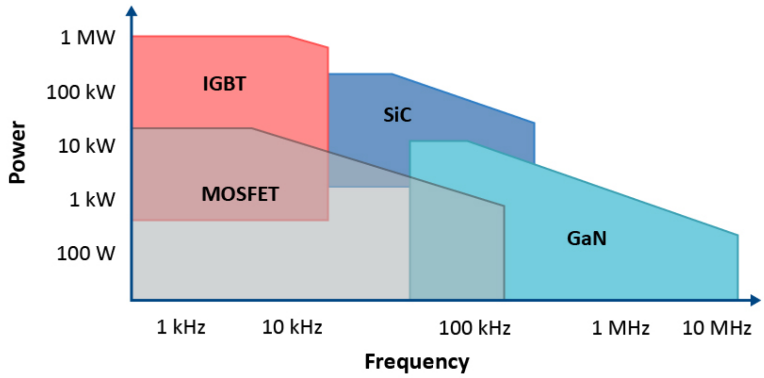

Figure 4.3: Technologies selection based on power and switching frequency [10].

4.2 Processing unit

4.2.1 MCU selection

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A Microcontroller Unit (MCU) typically consists of three fundamental components: a central processing unit (CPU), memory, and peripheral modules. The CPU executes program instructions and performs arithmetic and logical operations. Memory is generally divided into non-volatile memory (Flash) for program storage and volatile memory (SRAM) for data and runtime variables. Peripheral modules provide interfaces to the external environment and may include analog-to-digital converters (ADCs), timers, pulse-width modulation (PWM) units, communication interfaces (such as UART, SPI, I²C and CAN), and general-purpose input/output (GPIO) ports. The integration of these components enables standalone operation without the need for extensive external circuitry [32].

When selecting a MCU for a specific application, the first factor to consider is to use an 8, 16, or 32 bit MCU. For this application, a 16-bit MCU would probably be sufficient, however to ensure future-proof and compatibility with other Técnico Solar Boat (TSB) systems, the choice is for a 32-bit MCU. Besides, nowadays, 32-bit MCUs are considered the industry standard since most embedded system Research and Development (R&D) effort is focused on 32-bit cores, and thus both architectures can achieve similar power consumption [33]. There are two main architectures for 32-bit MCUs: AVR and ARM. Being ARM, the most widely used architecture in the industry and the most advanced, it was chosen for this project [34].

STM32 family of MCUs offers a wide range of solutions with different characteristics, making it suitable for various applications, including this project. This family of microcontrolers is used by TSB, so it will be used for this project to maintain the standard.

The MCU requirements for this project are:

- Fast clock frequency: 80 MHz or higher;
- Support 1x CAN peripheral, 1x I2C, 1x UART;
- Able to read 4 or more analog signals;
- 12bit ADC or more.

The final decision was the STM32L476. Although it is not the most suitable STM32 microcontroller for this application, it was selected due to its low cost and because it was already widely used within TSB, reducing development time and associated risks.

The STM32L476 is based on an ARM Cortex-M4 core with floating-point support and operates at frequencies up to 80 MHz. The device provides 1 MB of Flash memory and 128 kB of SRAM, which is more than sufficient for the implementation of the MPPT algorithm, data acquisition, communication protocols, and debugging features. Regarding peripherals, it includes three 12-bit ADCs with up to 16 external channels each, multiple general-purpose and advanced timers capable of high-resolution PWM generation, several communication interfaces (CAN, SPI, I^2C, UART, USB), and DMA controllers that allow peripheral operation with minimal CPU intervention.

These features make the STM32L476 particularly attractive for power electronics applications, where simultaneous voltage and current measurements, fast PWM generation, and real-time control execution are required. Furthermore, the presence of a hardware floating-point unit simplifies the implementation of control algorithms and numerical calculations.

If there were no limitations, a microcontroller with a higher clock frequency would have been selected, increasing both the PWM duty-cycle resolution and the available computational power. Additionally, a device with native 16-bit ADCs would improve measurement accuracy and reduce quantization effects. Nevertheless, the STM32L476 provides sufficient performance and peripheral resources to meet the requirements of this application.

4.2.2 Configuration

Since the designed board is custom-made, there is any pre-configuration of the MCU in use. In this particular case, the STM32 project needs to be configured in a STM32CubeMX, which is a tool that simplifies the configuration process. In this tool it is defined the Clock frequency, Peripherals working frequency, pin-maping (GPIOs, ADCs, UART, CAN, etc.) and respective configuration, Interrups, Timers, PWMs, etc. This tool generates a project base on C programming language with all drivers and configurations needed. These files serve as based for firmware development, and can be modified during the project.

Since it is not possible to describe all the configurations used lets highlight the most important ones for this project.

The MCU works at 80 MHz, with an external ceramic resonator of 16 MHz. This frequency is enough for the computational time, and a minimal PWM resolution. It can be flashed using JTAG and it uses an Independent Watchdog (IWDG) time interrupt. The IWDG is a safety feature that resets the MCU in case of a crash or infinite loops. It works by resetting the timer periodically in a specific part of the code, if the timer is not reseted during a specified period, the interrupt is triaged and the MCU resets.

As for timers, one timer was configured with 100 KHz for PWM and two pins were configured as complementary PWMs with a specific dead time. The dead time was inserted as a safety feature too. The dead time is calculated using Formula 4.26, where the ”value” needs to be in the range from 0 to 255.

$$DeadTime = Value/F_{CLK} \tag{4.26}$$

For this project it was tested several values using an oscilloscope. The smallest the value that ensures that both signals are never high at the same time is 2, however 3 was used to add margin. The measured dead time was 32ns, and the rising and falling time were 19ns and 22ns respectively (Figure 4.26).

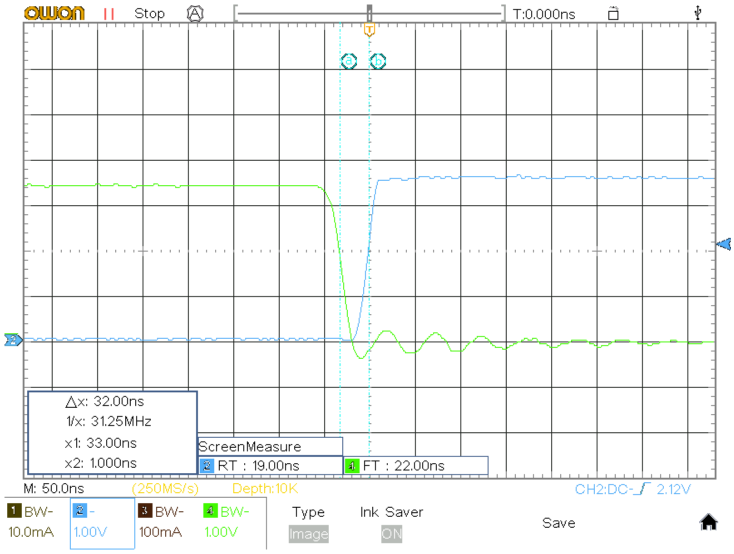

Figure 4.4: Dead time measurement with oscilloscope (value=3).

In simulations the dead time values of 2 and 3 were compared and the difference in efficiency is only 0.01% and therefore the margin is justified.

With 100 KHz PWM and 80 MHz internal clock, the dutty cycle resolution is 0.125 %.

Another timer was configured to trigger every millisecond, and it is used to serve as task interrupts however each task is called at difrent periods (always around milliseconds) which is implemented with a counter and comparators.

ADC config (SCAN mode + DMA), sample rate, resolution, mode

4.3 Sensors

4.3.1 Voltage and current sensing

Accurate voltage and current measurements are essential for MPPT operation, since tracking algorithms directly rely on these quantities to estimate power and adjust the operating point of the Photovoltaic (PV) array. Measurement noise or inaccuracies directly impact tracking efficiency, especially in high frequency switching DC-DC converters.

Voltage sensing is generally straightforward and can be implemented using resistive voltage dividers followed by ADC. Additionally, it is suggested to use an op-amp as a buffer to reduce the impact of the ADC input capacitor, while charging. This buffer reduces noise and improves the accuracy of the measurement while also allowing the use of higher sampling rates.

Current sensing, however, is more critical and requires careful consideration of the available measurement techniques.

The two main techniques are:

- Hall effect current sensor.
- Shunt resistor sensing with operational amplifiers;

The hall effect current sensor is a non-invasive method that uses the Hall effect to measure current. It provides galvanic isolation and can do bidirectional measurements. However, it is typically more expensive, has lower bandwidth, and may introduce additional noise due to its magnetic sensing nature.

As for the shunt resistor sensing, it is a simple and cost-effective method that uses a low-value resistor to convert the current into a voltage drop. This voltage is then amplified using operational amplifiers to match the ADC input range. This method provides high accuracy, low cost, and fast response time, but it requires careful design to minimize power loss since it will dissipate power proportional to the current.

In the end product a IC based on a shunt resistor sensing method was used, which as a built-in amplifier. The sensor chosen was the INA253A2IPWR, which has a PWM rejection block and small form factor that will be extremely useful for this application. However, this IC is more expensive than a simple shunt resistor and op-amp solution, but it is more accurate and has a better noise rejection. In the future it can be swapped for a lower cost solution, if the performance is acceptable.

Both sensors use the internal ADC from the STM32L476, which has a 12 bit ADC (with 3.0V as the reference). The current sensor output an analog voltage that is equal to the current multiplied by 0.2 V/A. Which means that the resolution of the current measured is 3.0/(4096 * 0.2) = 3.66 mA. As for the voltage sensor, it outputs an analog voltage that is equal to the voltage multiplied by 10/210. The resolution is 21 * 3.0/(4096) = 15.38 mV, which is more than enough for this application.

Both circuits are represented in Appendix 5.3. place the apendix reference

4.3.2 Temperature and Irradiance sensors

For security reasons it is important to monitor the temperature of the Printed Circuit Board (PCB). Also, the temperature and irradiance of the PV panel are important to monitor for performance analysis and defect detection. These measurements can also be used to improve the MPPT algorithm, since the Maximum Power Point (MPP) is dependent on temperature and irradiance.

For temperature sensing there is a wide range of options, from simple thermistors to more complex digital temperature sensors. For this application a simple thermistor was used, since it is cheap and easy to implement. The thermistor is connected to a voltage divider and the voltage is read by the ADC. The thermistor used to measure ambient temperature and transistor temperature was the ERT-J1VS104FA, which has a resistance of 100kΩ at 25 ℃ and a B value of 4330K. As for the thermistor used to measure the PV panel temperature, it was the 104JT-025, which has a resistance of 100kΩ at 25 ℃ and a B value of 4390K. The temperature can be calculated using the Steinhart-Hart equation [35].

$$\frac{1}{T} = A + B \cdot ln(R) + C \cdot ln(R)^3 \tag{4.27}$$

This equation uses 3 parameters (A, B and C) that are usually not provided in data sheet but can be calculated using 3 measurements of temperature and resistance. However, the most common approach is to use a simplified version of the equation, which uses only 2 parameters (B and R0). The temperature can be calculated using the following equation:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \cdot ln(\frac{R}{R_0}) \tag{4.28}$$

The thermistor that measures the transistor temperature is also used in an analog protection. This protection uses a comparator to disable the transistors after a temperature threshold that can be modified by changing on resistor in the comparator circuit.

As for the irradiance sensor, the TSL2591 was used, which is a digital light sensor with a wide dynamic range and high sensitivity. It can measure light intensity from 0.000118 lux to 88,000 lux, making it suitable for various lighting conditions. The sensor communicates with the MCU via the I2C protocol, allowing for easy integration and data acquisition.

Since it measures the light intensity in lux, it is not a direct measurement of irradiance and it needs to be calibrated to convert the lux value to irradiance in W/m$\hat{2}$. This calibration was done using a irradiance sensor from ??? which measures the irradiance in W/m$\hat{2}$. The calibration was done by measuring the lux and irradiance at the same time and calculating the conversion factor. The conversion factor was found to be ???? W/m$\hat{2}$ per lux at laboratory conditions. This factor changes with the spectral distribution of the light source, so it would need be calibrated for different light sources.

There are bether options for irradiance sensors, but this sensor was chosen due to its low cost and easy integration with the MCU. The main disadvantage of this sensor is that it is not a direct measurement of irradiance and it needs to be calibrated. However, for laboratory testes it is sufficient since the objective is to record when does the irradiance change to analyze the performance of the MPPT algorithm. For precise measurements of irradiance, the sensor used to calibrate the TSL2591 can be used, which is a more expensive sensor that provides direct measurements of irradiance but does not provide communication protocol to record the data.

In the future, if the irradiance value is needed to be used in the MPPT algorithm, a more precise irradiance sensor is needed. The most precise sensors use a PV cell, which based on his temperature, short circuit current and open circuit voltage can provide a good estimation of the irradiance. My suggestion is to design this circuit and use the same cell that is used in the PV panel to measure the irradiance. This way, the irradiance measurement will be accurate and with exactly the same spectral distribution as the PV panel which will increase the performance of the MPPT algorithm. This was not implemented in this project due to time constraints, but it is a good option for future work.

4.4 Software

In this section it is explained how does the firmware inside the MCU is structured and how it works. Also, the basic idea behind the communication protocols used and how they are applied to this project as well as the Graphical User Interface (GUI) developed for configuration and data analysis.

4.4.1 Firmware Overview

The firmware is a flag based scheduler with two main task, the MPPT and ADC task.

The ADC task is responsible to convert the analog readings into another units such as current, voltage and temperature. If any of the values exceed the limits configured by the user, a fault is detected and blocks the MPPT to run during 1 second. After this time the ADC task runs again and checks if the faults was removed.

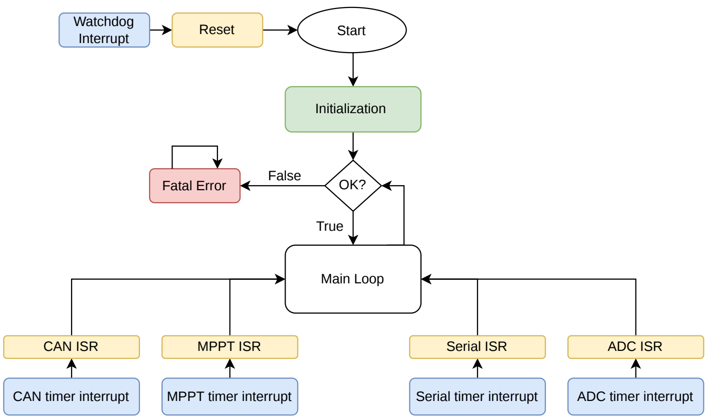

Figure 4.5: Firmware flowchart.

The MPPT task is responsible to run the state machine which changes the operation mode depending on the measurements and outputs a PWM duty cycle value. In Figure 4.6, is represented how does the code transit between states. The idea is, if the board is not connected to a battery, the output voltage is set to the Constant Voltage (CV) target (No load state), then if enough current is detected in the battery side it switches to Sweep mode (if enabled) which sweeps the PV operating point to find the MPP. Once the sweep is finished it changes to the CC-CV charging phase, where depending on the battery current and voltage, the mode will be MPPT algorithm, Constant Current (CC) or CV. Once the battery finish the charging cycle or disconnects from the board, it changes to No load mode again. When designing these state machine, one of the most important aspects was the ability of moving forward in terms of charge (first CC then CV) or backwards in cases where the battery discharge current is higher in comparison with the energy provided by the solar panels.

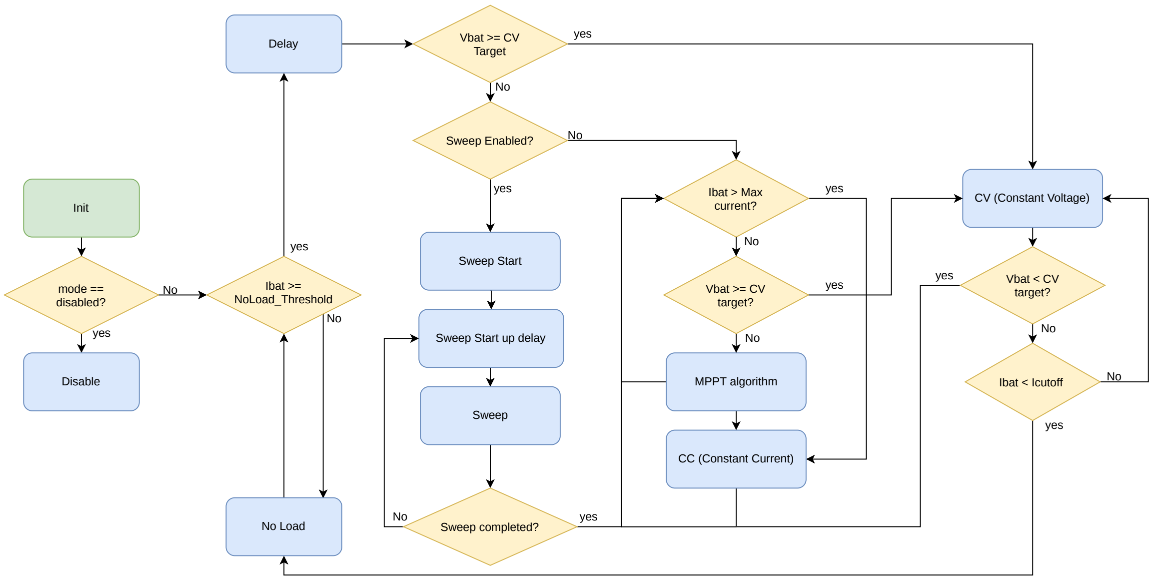

Figure 4.6: Charging state machine state.

Besides the main tasks, the board is able to send CAN and Serial messages and therefore the respective tasks.

4.4.2 Serial and CAN communication

One of the most important features of this circuit board is the communication with the rest of the boat. This will enable better decision making relating to energy budget. Also, it will be easier to detect broken or malfunction solar panels or human errors such as bad or forgotten connections.

The main communication protocol is Controller Area Network (CAN). This is the TSB standard protocol and widely use in automotive industry due to its high robustness. The protocol was developed by Bosh in 1986 to let microcontrollers and devices talk to each other without a central host computer.

In the hardware point of view, CAN uses a bus instead of wire to wire communication. Also, it uses twisted wires (CAN High and CAN Low) to block electrical noise. In a software point of view, it uses an error checking system (CRC) ensuring reliability.

For TSB electrical system we developed our CAN protocol on top of the standard CAN communication. Each message is structured as shown in Figure 5.3 following the CAN 2.0A standard.

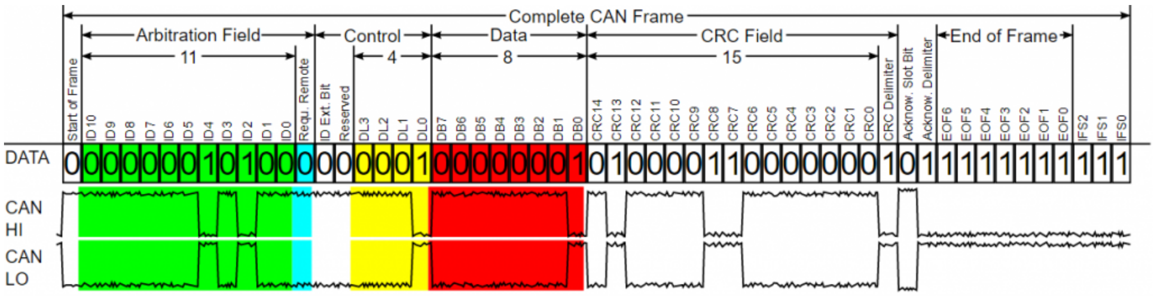

Figure 4.7: Dashboard.

The 11 bits marked in green (Arbitration Field) can be used by each protocol differently. Usually, it is used to send priority bits, device ID and message ID. In the current vessel (São Gabriel 01 (SG01)), the priority bits were removed and used to increase the number of devices on bus. With the current structure Tab 4.3, the protocol supports up to 32 devices with 63 messages each.

Table 4.3: TSB CAN messages Arbitration Field.

Message ID	Device ID
6 bits	5 bits

In Control field, the first bit is to use Extended IDs and next one is reserved. The yellow field is called Data Length Code (DLC) and is where the length of the data is defined. The maximum lenght is 8 bytes. Then there is the Data field (represented in red).

The rest of the fields are treated by the CAN library which is an adaptation of the open source library called `STM32_CAN` by pazi88.

For the MPPT it was defined an ID structure, where the device ID is maintained intact, and Message ID is divided in two. The 3 most significant bits represent the MPPT number from 0 to 7 and the rest of the bits identifies the messages. This structure gets enough message IDs for each MPPT as well as not occupying a lot of Device IDs. In Table 4.4 is showed an example of the corresponding IDs when the Device ID is 0x0D (default ID).

The Device ID can be configured/changed via GUI however the logic is maintained.

The CAN protocol is excellent for communication between boards however it is not the most practical protocol for configuration because the CAN IDs can change between vessels and implies the use of an external hardware to communicate between the board and a computer.

In contrast, serial communication is done by just connecting a wire to the board and since the protocol is only used for configuration and data analysis it can be the same forever and if it needs to be changed, it does not depend on any vessel.

Table 4.4: CAN IDs for each MPPT number and message.

MPPT Number	MSG:MPPT_bat	MSG:MPPT_pv	MSG:MPPT_aux1	MSG:MPPT_aux2
0	0x00D	0x02D	0x04D	0x06D
1	0x10D	0x12D	0x14D	0x16D
2	0x20D	0x22D	0x24D	0x26D
3	0x30D	0x32D	0x34D	0x36D
4	0x40D	0x42D	0x44D	0x46D
5	0x50D	0x52D	0x54D	0x56D
6	0x60D	0x62D	0x64D	0x66D
7	0x70D	0x72D	0x74D	0x76D

There is no default protocol to send data over serial, however in Battery Management System (BMS) thesis a serial protocol was created and used and will serve has based for this project. Unfortunately the firmware library is not compatible with STMs, so it was created from scratch.

The protocol uses the structure defined in Table 4.5. It begins with the Start Byte which is defined as 0xAC, followed by the device ID which defines the source of the message and the message ID defining what message is the device sending, the data length, the data, checksum byte and the end byte (0xAD) marking the end of the message.

Table 4.5: TSB Serial messages format.

Start	Device ID	MSG ID	LEN	Data	Checksum	END
1 Byte	1 Byte	1 Byte	1 Byte	8 Bytes	1 Byte	1 Byte

Similar to CAN protocol, for each message it was defined a Message ID respecting the same logic. This ID alongside with the Device ID and length is initialized in the beginning of the code, the rest of the variable are treated by the library except the data field that is changed before sending any message.

The Data of each message is structured in the same way as in **CAN!** (**CAN!**) protocol however it was added 4 messages for configuration. These messages enable the change of current limits, voltage limits, MPPT parameters, CAN settings and CC-CV targets.

4.4.3 EEPROM Emolation

To save the configuration received from the serial protocol, a library was used that emulates an EEPROM which ia a type of non-volatile memory that retains its contents after power-down. EEPROMs have been used for many years in a wide range of devices to store settings, calibration data, and backups. In this project, no external EEPROM was added to the board, and the MCU does not include one internally. However, the library can use internal flash memory, which is normally reserved for program code, to store data, thereby emulating EEPROM behavior.

The main disadvantages of this approach are limited storage capacity and reduced write endurance compared with a dedicated EEPROM. The STM32 internal flash can typically withstand on the order of 10,000 erase cycles per page, whereas external EEPROM devices often support roughly one order of magnitude more. Even so, this is not a practical limitation in this application. Configuration data is stored as a small set of 16-bit variables, and each save operation appends new records to flash rather than overwriting the previous value in place. Each page can hold 252 variables. For this project, a full board configuration requires 13 write operations, therefore, one flash page can hold many configuration updates before it becomes full and must be erased. Furthermore, the emulation is configured to use ten flash pages, with wear levelling implemented through page transfer and cleanup. When a page is full, the latest values are copied to another page and the old page is eventually erased. This distributes erase cycles across multiple pages and extends the effective lifetime of the memory.

In conclusion, neither storage capacity nor endurance is expected to be a problem for this project. Under the current configuration, a complete configuration upload would need to be performed hundreds of thousands of times before the theoretical endurance limit of a flash page is reached—far beyond the expected use case of occasional configuration updates over serial.

Since the emulated EEPROM shares the same physical flash as the firmware, care must be taken to ensure that program code is never placed in the reserved EEPROM region. This was achieved by modifying the linker script, which defines the memory map used during the build process. The length of the program flash region was reduced so that firmware is linked only below the first address saved for EEPROM emulation.

4.4.4 Graphical User Interface

In parallel with the design and test of the MPPT board, a GUI was designed to make this project more accessible and easy to use. The application was built from scratch, however it was inspired in the application built during BMS thesis. It was built using QT framework, which is a powerful, cross platform software toolkit used to build high-performance applications and GUIs that run on multiple operating systems like Windows, macOS, Linux.

This GUI communicates with the board with Serial or CAN communication. To use CAN communication, the computer should be connected to the bus with a tool called PEAK-CAN.

After connecting the computer to the board, the application is able to identified which board is connected and show it to the user in the Dashboard page, Figure 4.8. In this page the user selects which board he is working on. The application is prepared for several PCBs, however only the BMS and MPPT are correctly working. The rest are placeholders for future improvements.

When the MPPT page is selected (Figure 4.9), a new tab opens showing the most important variables, such as voltage, current, power temperature, efficiency, irradiance and duty cycle. Additionally, there are two visual representations. On the left there is a P-V graph with theoretical curves for different irradiance values and a blue dot placed on the current point of operation and updated with the received

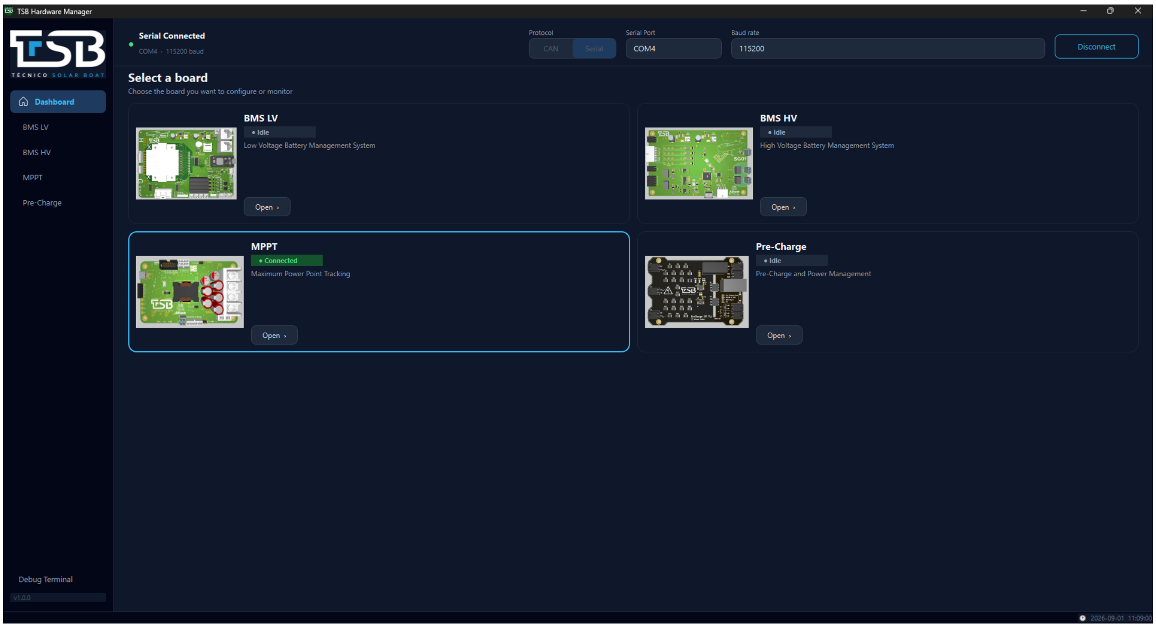

Figure 4.8: Dashboard page.

data.

On the right, it is represented the CC-CV curve, where the received battery side data is represented over time.

Figure 4.9: MPPT tab.

On top of the page, the user can switch to the configuration tab (Figure 4.10). In this tab the user can read the configuration currently installed on the board, reset this configuration or upload a new one.

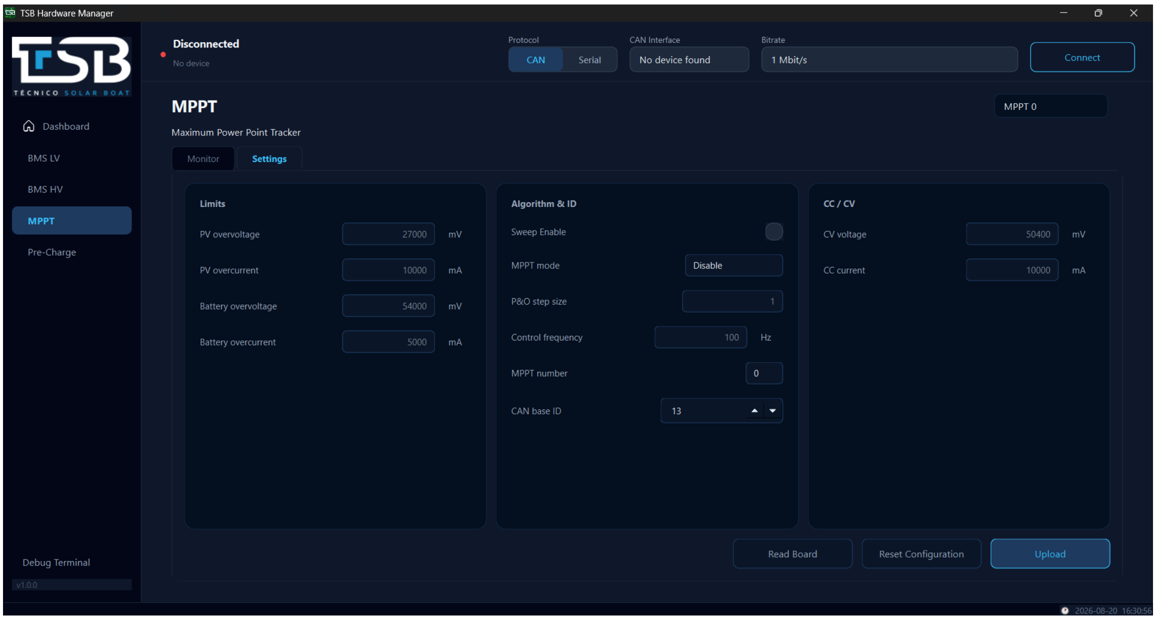

Figure 4.10: MPPT configuration tab.

The last page is the debug page (Figure 4.11), where the user can visualize both Serial protocol messages and serial prints at the same time.

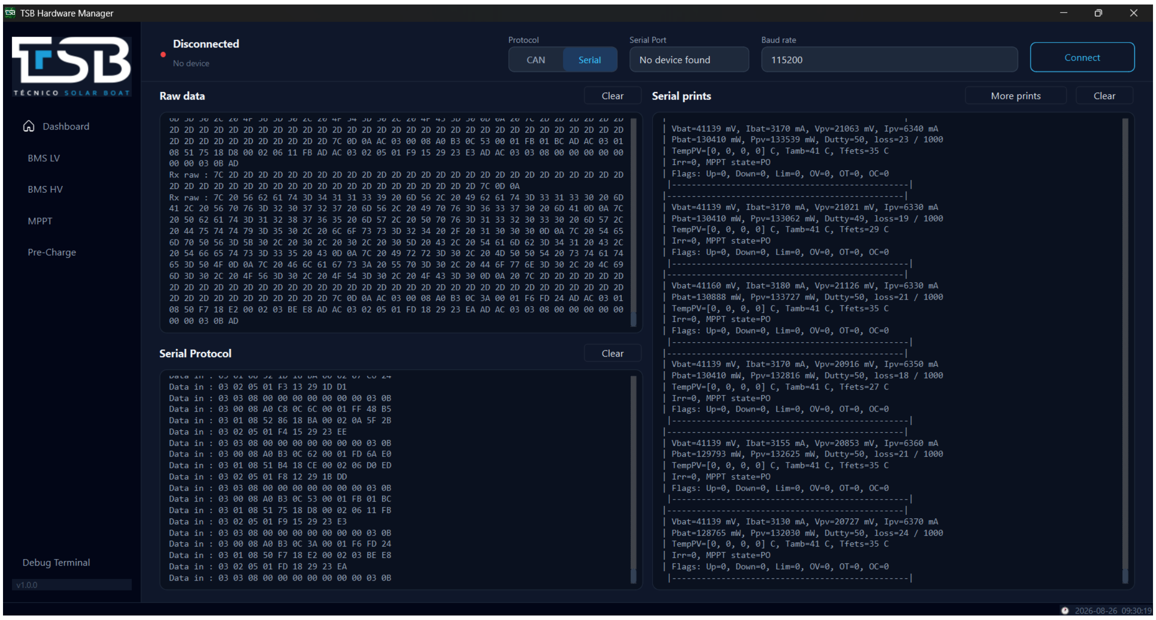

Figure 4.11: Debug page.

5

Simulations

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Objective

5.1 DC-DC Converter

5.1.1 Solar Panel model

The simulations started at LTspice with the aim to test the converter at different conditions. To achieve that, a Photovoltaic (PV) electrical model of the cells were designed using a five parameter model (1M5P) configuration. There are more electrical models that can simulate a solar panel, some with more accuracy but more complex. This model was perfect for the aim of the simulation since it is both reliable and simple.

The 1M5P model represents the solar cell through a single diode equivalent circuit composed of a current source in parallel with one diode, a series resistance R_s, and a shunt (parallel) resistance R_{sh}, as shown in Figure 5.1. The current source represents the photogenerated current I_{ph}, which is proportional to the incident irradiance, while the diode accounts for the non linear I-V characteristic of the p-n junction. The series resistance models the losses associated with the material bulk resistance, metal contacts and interconnections, whereas the shunt resistance represents the leakage currents through the cell, typically caused by manufacturing defects.

The output current of the model is given by the characteristic equation 5.1, where I_0 is the diode saturation current, n is the diode ideality factor, and V_t is the thermal voltage [36]. Together with I_{ph}, R_s and R_{sh}, these constitute the five parameters that give the model its name.

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \tag{5.1}$$

These parameters are not directly provided by manufacturers, but can be extracted from the standard datasheet values, namely the open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), voltage and current at the Maximum Power Point (MPP) (V_{mp}, I_{mp}), and the respective temperature coefficients, through a set of equations that ensure the model matches these reference points under Standard Test Conditions (STC). There are several ways to do it, however it is outside the range of this work. To simplify, calculations were carried using online tools and will not be explained in detailed here. Se achar necessario posso escrever as equações aqui!?!?

This approach allows the panel behavior to be accurately reproduced under different irradiance and temperature conditions, while keeping the circuit implementation simple enough to be integrated directly into LTspice.

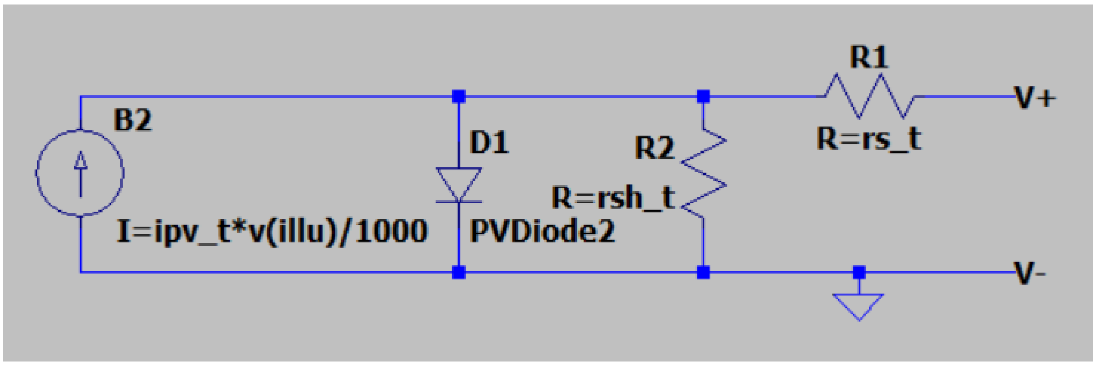

Figure 5.1: Solar panel model diagram (1M5P configuration).

5.1.2 Boost Converter

To validate the theoretical calculations, simulations in LTspice were carried using the boost converter in steady state mode and a fix duty cycle.

Considering a solar panel with 35 cells, five points of CC-CV charging curve were chosen to test the performance of the converter. These five points are represented in Figure 5.2.

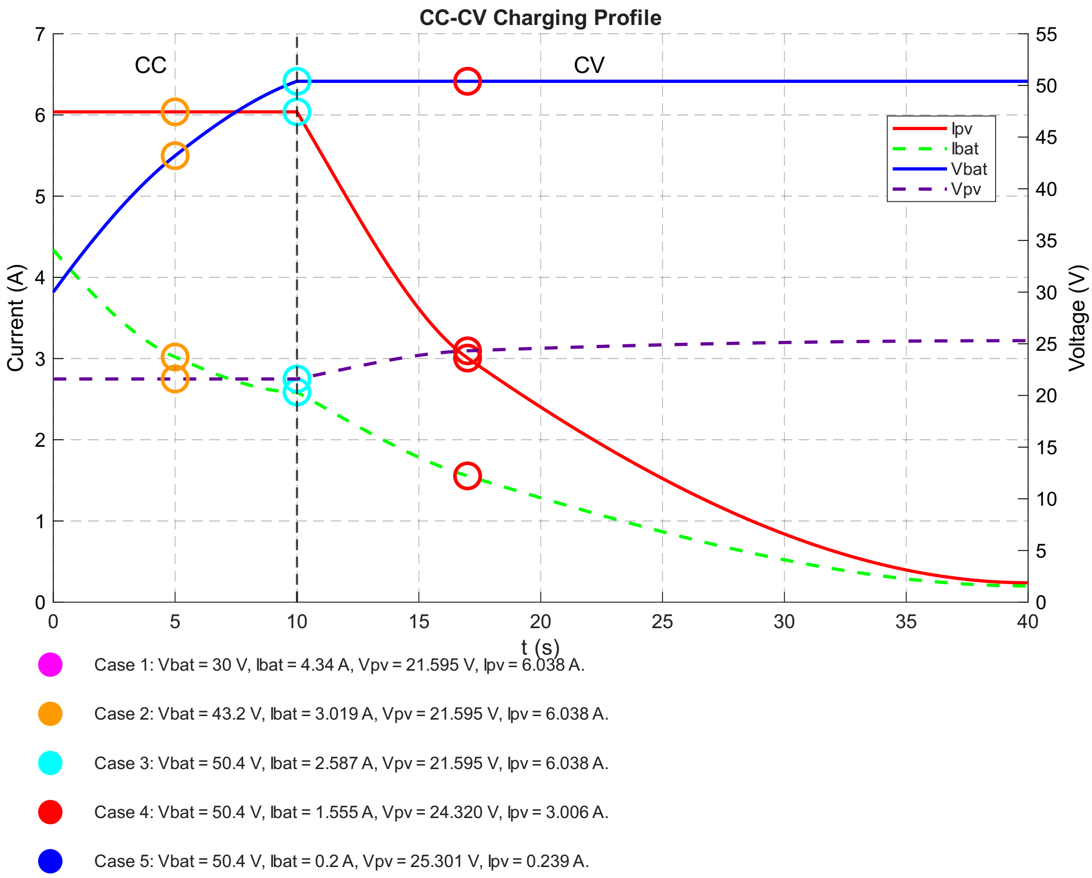

Figure 5.2: CC-CV charge characteristics and analysis point selection.

It is important to note that in this CC-CV curve the constant current is given by the solar panel and not by the battery as it is usually expected. This is only truth due to the maximum charge current of the battery being lower than the current that can be provided by the solar panel. So at CC, the converter is working at MPP.

With this five points chosen, some simulations were carried using capacitors and inductors that would fit the requirements calculated earlier. The objective of these simulations is to optimize the performance of the converter, leading to better choices.

The model was built in LTspice with approximated models given by the manufactures for higher precision. Only the solar panel model was designed by hand, as previously mentioned.

The battery was simulated using a voltage source and a resistor representing the internal resistance of the battery.

The dead time of the transistors was implemented too. Using laboratory measurements, the rising and falling timing was measured and used in simulation. Has for the dead time, it is used the same as in the final prototype.

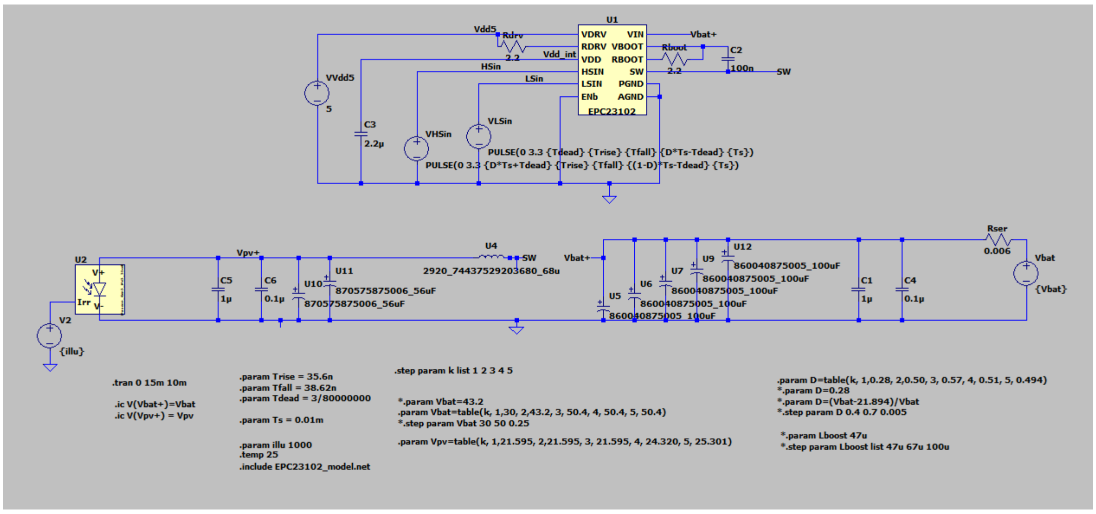

Figure 5.3: Simulation circuit schematic.

The first simulations were carried to choose the input and output capacitors. The idea is to simulate the circuit using a known inductor (68 uH) and change the capacitors for different configurations. All configurations are overdimensioned to give low ripples (current and voltage). So the main factor is the efficiency of the converter.

In Table 5.1 is shown the results. The efficiency of all combinations are very high however the highest is the one with two 56 uF as input capacitors and tree 220 uF capacitors as output capacitors. Unfortunately the 220 uF were not available at the moment, so the second best option was chosen, the five 100 uF capacitors.

It is worth noting that this is a simulation and therefore the efficiency is higher than what will be measured in laboratory results. Additionally, the difference between each combination it is very small. To make the product cheaper the third option would be as great as the one chosen.

Table 5.1: Loss (%) for different capacitor configurations (Cin, Cout).

Case #	2x56 μF, 3x220 μF	100 μF, 3x220 μF	100 μF, 5x100 μF	2x56 μF, 5x100 μF
1	0.5729	0.5796	0.5856	0.5788
2	0.6762	0.6985	0.7069	0.6844
3	0.7359	0.7655	0.7743	0.7449
4	0.6654	0.7091	0.7161	0.6735
5	3.4177	3.7148	3.7554	3.4671

Similar to the previous simulation, the inductor was chosen by keeping all components with the same values (the capacitors used were Cin = 2x56uF and Cout = 5x100uF) and changing the inductor. Similar to before, all inductors simulated are within the minimum required.

The losses for each case are represented in Table 5.2. Similar to before, the efficiency difference is very low but not as linear as in the previous simulation, likely due to the inductor models chosen.

The inductor chosen will highly influence the input ripple which can generate heat, and electrical interference. Table 5.3 shows the different ripple currents for each inductor in each case of the CC-CV curve. The 22 uF inductor showed higher ripple current reaching values close to the input current. The inductor chosen was the 68 uF since it has overall good efficiency and reasonable input current ripple.

Table 5.2: Loss (%) for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	0.4206	0.4439	0.5453	0.5788	1.2659
2	0.6645	0.6793	0.6627	0.6844	1.3664
3	0.7883	0.8076	0.7290	0.7449	1.4193
4	1.0556	1.0583	0.6867	0.6735	0.9624
5	6.7045	6.6230	3.6556	3.4671	3.8177

Table 5.3: Δi_{in} $(I_{in,\max} - I_{in,\min})$ for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	2.6939	1.8906	1.3256	0.9781	0.6251
2	4.7934	3.3416	2.3354	1.7066	1.0707
3	5.4733	3.8253	2.6813	1.9644	1.2839
4	5.5596	3.8565	2.7075	2.0139	1.2925
5	5.5520	3.8510	2.6903	1.9636	1.2344

Now that the converter components are all chosen, the following measurements were taken in LT-spice:

Iin é difrente de IL. O Iin tem quase nada de ripple

Both voltage ripples are lower than what was defined as the goal in Table 4.2. As for the current ripple it is still reasonable. The losses/efficiency are also higher than what was aimed, reaching values

Table 5.4: Ripple and Power Loss Summary for Each Operating Step

Case #	ΔV_{bat} (V)	ΔV_{pv} (V)	ΔI_{in} (A)	Loss (%)
1	0.0406	0.0145	0.9781	0.578
2	0.0491	0.0212	1.7066	0.684
3	0.0529	0.0244	1.9644	0.745
4	0.0406	0.0220	2.0139	0.674
5	0.0302	0.0215	1.9636	3.464

up to 99.4%, in simulation.

A voltage sweep were performed, and as expected the efficiency drop with the increase of output voltage (Figure 5.4). Also, as mentioned earlier, the output current drop because in Constant Current (CC) mode the solar panel current is constant, so the output current drops with the increase of battery voltage due to the power conservation principle.

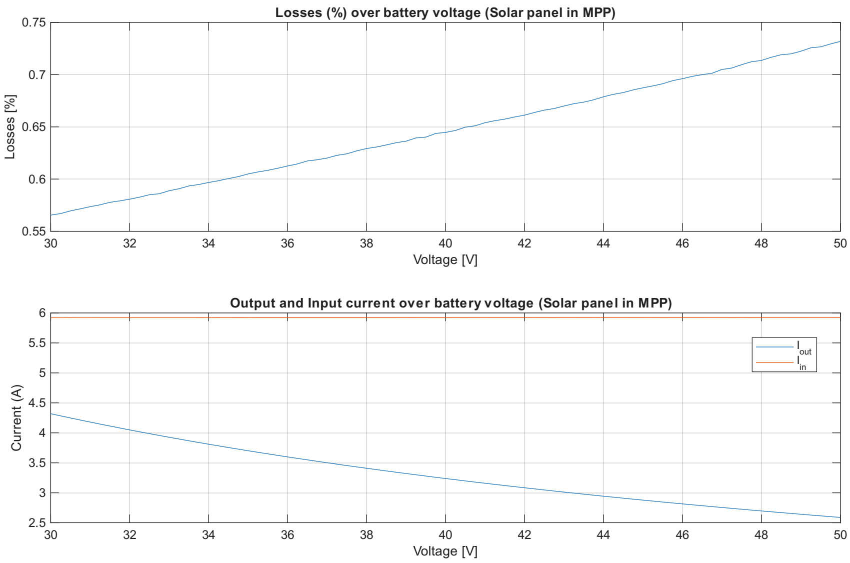

Figure 5.4: Battery voltage sweep simulation results.

Note: in Section ?? it is mention the addition of two shotcky diodes that were not considered in simulations. These diodes slightly increase the efficiency of the converter (+0.03%) due to their lower forward voltage.

Add wave forms??!?!?!

5.2 Power Loss Estimation

Although LTspice provides an accurate estimation of the converter efficiency, it does not directly identify the contribution of each component to the total power dissipation. Therefore, an analytical power loss model was developed in MATLAB to estimate the individual losses of each component of the proposed synchronous boost converter. The model uses the electrical parameters of the selected components obtained from their datasheets together with the converter operating conditions.

The total converter losses are obtained by summing the losses associated with the GaN transistors, inductor, capacitors, dead time and gate driver,

$$P_{loss} = P_{cond_{HS}} + P_{cond_{LS}} + P_{sw_{HS}} + P_{sw_{LS}} + P_L + P_C + P_{dead} + P_{gate}. \tag{5.2}$$

The converter efficiency is then calculated as

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100\%. \tag{5.3}$$

The conduction losses of each GaN transistor are given by

$$P_{cond} = I_{RMS}^2 R_{DS(on)}. \tag{5.4}$$

These losses are primarily caused by the on state resistance of the transistors, resulting in power dissipation whenever current flows through the channel.

The switching losses are estimated using Equation 5.5, which takes into account the loses during the period when the transistor is not fully nor fully off. Also, the losses associated with charging and discharging the output capacitance of each transistor.

$$P_{sw} = \frac{1}{2} V_{out} I_{L_{AVG}} (t_r + t_f) f_{sw} + \frac{1}{2} C_{oss} V_{out}^2 f_{sw} \tag{5.5}$$

The dead-time losses are calculated using Equation 5.6. These losses originate from the conduction of the reverse conduction path during the dead-time interval required to prevent shoot through. This reverse condition path is either from the third quadrant conduction from the GaN device or from the parallel schottky diode added later in version 1.1 (this is explained in Section ?)

$$P_{dead} = 2 \cdot V_f \cdot I_{L_{AVG}} \cdot t_{dead} \cdot f_{sw}. \tag{5.6}$$

The inductor and capacitor losses are estimated using the RMS current passing through them and their Equivalent Series Resistance (ESR)

$$P_L = I_{L_{RMS}}^2 ESR. \tag{5.7}$$

$$P_C = I_{C_{RMS}}^2 ESR, \tag{5.8}$$

Although magnetic core losses of the inductor could be estimated too, the selected manufacturer does not provide the required magnetic material parameters. Consequently, only the winding copper losses are considered, leading to a slight underestimation of the total converter losses.

reler o texto, inserir as perdas do circuito auxiliar, inserir as perdas vindas do REDEEXPERT, adicionar as perdas vinda do sensor de corrente, simplificar ou não a pie chart

Finally, the gate driver losses are estimated using Equation 5.9 which takes into account the energy needed to drive the transistors. Although the EPC23102 has an internal gate driver, this is still relevant.

$$P_{gate} = (Qg_{LS} + Qg_{HS}) * Vdrv * fsw. \tag{5.9}$$

Figure 5.5 presents the resulting power loss distribution. The calculated total power loss is approximately 1.188 W, corresponding to an efficiency of 99.10%, which agrees well with the LTspice simulation results presented in the previous section.

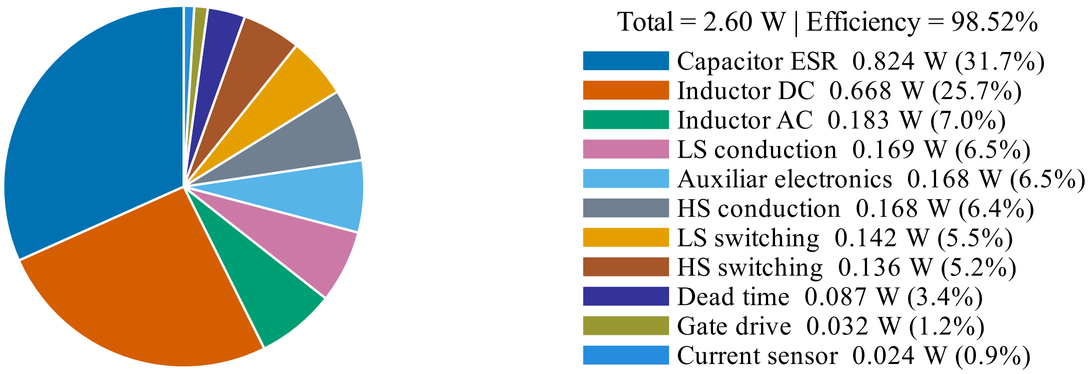

Figure 5.5: Analytical power loss distribution of the synchronous boost converter for Case 1. change for the case 2

The largest contributors to the total losses are the inductor copper losses and the capacitor ESR losses, each accounting for approximately 31.5% of the total power dissipation. This highlights the importance of selecting an inductor with low winding resistance and capacitors with low ESR, as these passive components dominate the converter losses.

The conduction losses of the GaN transistors account for approximately 16% of the total losses, showing the importance of choosing transistors with low conduction resistance. The high-side device dissipating more power than the low-side transistor due to its longer conduction interval. Switching losses represent approximately 12% of the total losses, including both transition losses and the energy required to charge and discharge the output capacitances. Dead-time losses contribute approximately 7%, while the gate driver losses are almost negligible (only 2.7%).

Additionally, all Printed Circuit Board (PCB) connections have an inductance that will represent losses too.

Overall, the analysis indicates that nearly two-thirds of the converter losses originate from the passive components, whereas the semiconductor devices account for approximately one-third. It should be noted that the presented values are based on analytical equations and datasheet parameters. Since magnetic core losses were not considered, the calculated efficiency represents a slight overestimation of the actual converter performance.

However, both simulation and theoretical analysis indicate that the choice of each components lead to a highly efficient converter.

5.3 MPPT algorithm

Simulik MPPT algorithm simulation.

6

Laboratory results

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Following the design and simulation stages, a prototype was manufactured and assembled. However, practical implementations may differ from simulation results, as simulations generally rely on idealized component models and do not fully account for parasitic effects associated with the components and the Printed Circuit Board (PCB). Therefore, laboratory testing was conducted to experimentally validate the design and verify that the prototype operates as expected.

6.1 Assembly and first iteration

All boards developed in this project were manufactured by Eurocircuits and assembled by hand in order to reduce costs.

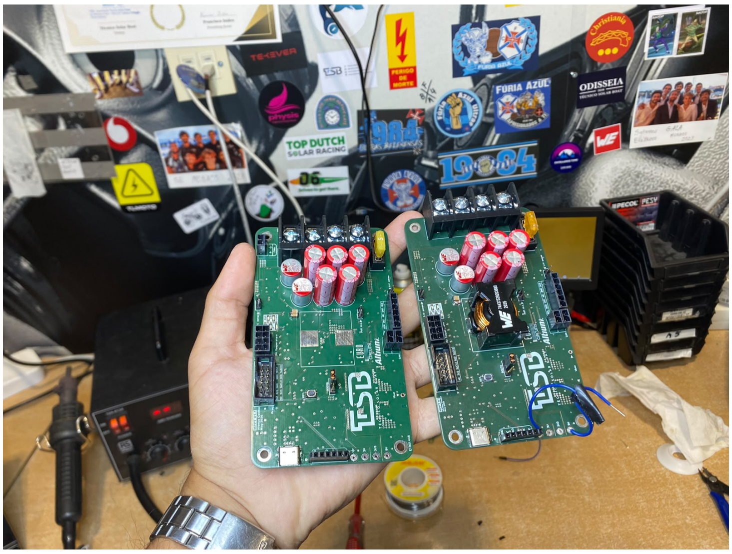

Figure 6.1: First and second iteration side by side. place a better image

The first prototype iteration presented several issues that were identified during the initial testing phase. For instance, an incorrect 3D model resulted in the main connector being positioned incorrectly (Figure 6.1). Additionally, the flash connector did not provide 5V which made test more difficult. The UART signals connected to the usb-c connector were swapped, and therefore the communication was only possible using a external USB to UART adapter.

Although none of these issues were critical for the fundamental operation of the circuit, they made the testing phase more time-consuming and complex. Therefore, a second iteration was designed and manufactured with all the issues fixed.

In the second iteration two schottky diodes were placed in parallel with the boost transistors which accordingly to the datasheet of the transistors, should improve the efficiency of the boost converter. This will be tested later in this section.

Furthermore, during testing of the first iteration, an increase in the inductor temperature was observed. To address the possibility of requiring additional cooling, a connector was added to place a fan if needed. These will also be analyzed in this section.

6.2 Voltage and Current sensor

Both current and voltage sensors were tested in the laboratory to validate the performance and accuracy of the measurements.

The voltage sensor test was performed by applying a known voltage to the input and measuring MCU readings.

The test was performed for the entire range of voltages needed for this application, which is from 0 V to 51 V. The results show that the voltage sensor is accurate and has an average error of 60 mV and 112 mV for battery and solar panel respectively. The error is different for each sensor because it depends on the actual resistance of the resistors used in the voltage dividers and the PCB layout. For each board the error will be different, but it is expected to be in the same order of magnitude since the tolerance of the resistors are equal.

These could be improved even further by using resistors with less tolerance and lower temperature coefficient or by increasing the ADC resolution. Another option that does not increase the cost of the system is to subtract the linear error of the results. The magnitude of the error is small, and therefore it does not affect the Maximum Power Point Tracker (MPPT) operation, however, for the purpose of this project, the linearization was not implemented in firmware to improve the accuracy of the measurements made during the testing phase.

In Figure 6.2 the error of the voltage readings is represented. Since the error is linear, the trend line is also represented, which was used to correct the results and improve the accuracy of the measurements. The results after subtracting the linear error are also represented in the figure, and it can be seen that the error is significantly reduced. The average error reduced to 28.325 mV and 29.825 mV for battery and solar panel respectively.

The same test was performed for the current sensors, the average error was 99.93 mA and 8.5 mA for battery and solar panel respectively, which is acceptable for this application. The error of the current sensors is shown in Figure 6.3, and it can be seen that the error is linear for the battery side and non-linear for the solar panel side. This can be explained by the fact that at low currents, the error is smaller and probably the shunt resistor value in the solar panel side is more accurate than the shunt resistor in the battery side. However, it is still very small and does not affect the MPPT operation.

As in the case of the voltage sensor, the subtraction of the linear error would highly increase the accuracy of the measurements. However, it takes a lot of time to do the measurements and calculate

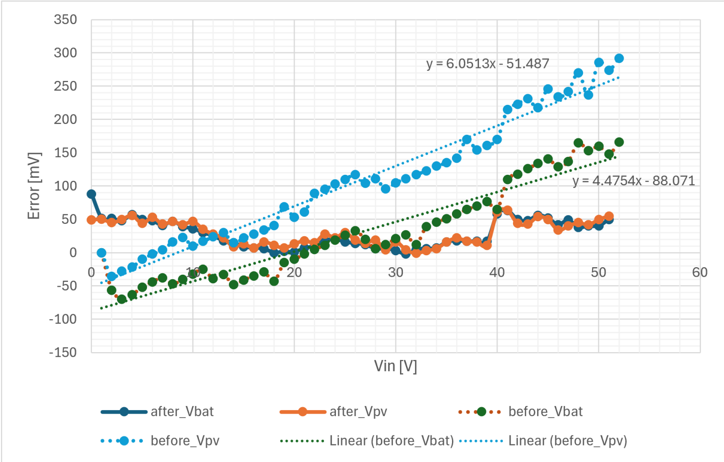

Figure 6.2: Voltage sensor error over voltage.

the linearization for each PCB which is not realistic for a student prototype project like Técnico Solar Boat (TSB). However, to increase the accuracy of the laboratory results, the error correction was implemented to the battery side current sensor. The results after subtracting the linear error are also represented in the figure, and it can be seen that the error is significantly reduced. The average error reduced to 5.28 mA.

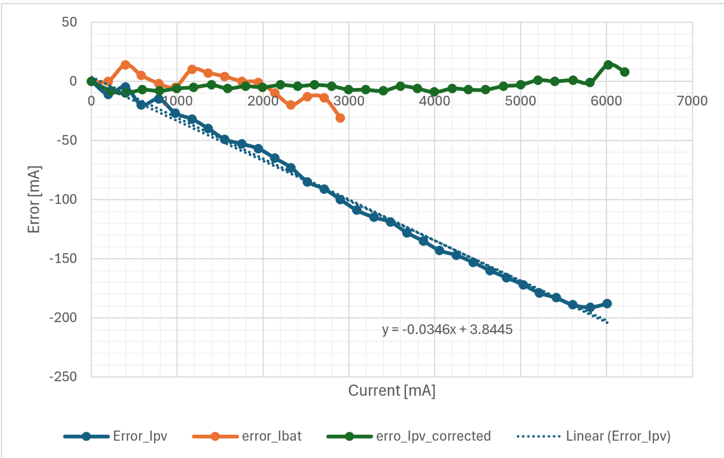

Figure 6.3: Current sensor error over current.

6.3 Boost

The Boost test is essential to identify the efficiency and electrical parameters of the board without the influence of the control system. Additionally, it is essential to confirm the correct dimensioning of the power converter enabling a comparison with simulations and theoretical calculations.

The test was performed using a voltage source on the input side and two power resistors on the load side. As in simulations, both duty cycle and load resistance were varied to obtain the results in all different points of the charging curve, Figure 5.2.

All the measurements were performed with multimeter with high precision. The voltage readings were performed as close as possible to the PCB, as for the current readings, they have to be performed in line.

It is important to note that this board uses the battery side to power itself, however in this test the battery side is not connected to a battery but since the Photovoltaic (PV) side is connected to a voltage source, this energy flows to the battery side through the boost converter diodes/body diodes and therefore powering the control system.

6.3.1 Switching noise

During the initial experimental tests, a significant amount of electrical noise was observed in all measured signals. The measured noise caused the current to exceed the configured Over Current (OC) protection thresholds, resulting in repeated activation of the protection mechanism. Initially, it was assumed that the problem originated from the PCB measurement system. Consequently, several modifications were implemented in the firmware, mainly increasing the ADC sampling time and increasing the number of samples acquired for each measurement, with the final measurement being obtained by averaging the collected samples. Furthermore, the fault-handling logic was modified because, once a OC fault occurred, the converter entered an oscillatory state in which it repeatedly enabled and disabled itself. Now when a fault occurred, it waits 1 second and verifies if the fault was removed. Although these software modifications improved the stability and accuracy of the measurements, they did not eliminate the excessive noise.

During the OC events, a large oscillation in the input current was also observed, varying approximately between 4 A and 12 A. One possible explanation considered was that the laboratory power supply was operating close to its current limit, causing it to become unstable. Replacing the power supply was therefore considered as a potential solution. However, since the observed noise was considerably higher than expected, it was believed that the root cause was located elsewhere.

A detailed analysis of the reference evaluation board revealed a significant difference in the output decoupling network. While the initial prototype employed a single 1 μF ceramic capacitor, which completely eliminated the switching noise in simulation, the reference evaluation board used ten 1 μF ceramic capacitors connected in parallel.

To evaluate the impact of the output capacitance, the converter was characterized without any changes. Figure 6.4 shows the initial noise.

In all oscilloscope captures, Channel 1 (yellow) corresponds to the output current (I_{out}), Channel 2 (blue) to the input voltage (V_{in}), Channel 3 (orange) to the input current (I_{in}), and Channel 4 (green) to the output voltage (V_{out}). Although the screenshots are not of optimal quality, they clearly illustrate the reduction in switching noise obtained after increasing the output decoupling capacitance.

To increase the local output decoupling, nine additional 1 μF ceramic capacitors were soldered in parallel with the existing capacitor, matching the total capacitance used in the evaluation board. After this modification, a significant reduction in the measured noise was observed, as shown in Figure 6.5.

These results confirm that the insufficient output decoupling capacitance was the primary contributor to the observed electrical noise. Increasing the number of ceramic capacitors substantially improved the converter performance and reduced measurement fluctuations.

Although the results were already satisfactory, the reference evaluation board also includes seven additional 220 nF ceramic capacitors distributed across the output stage. Initially the board also had one smaller capacitor (100 nF) and since it was not possible to get 220 nF with 100 V rating, a six 100 nF capacitors were added. The result should be similar. These capacitors are expected to further reduce the high frequency switching noise. The results are presented in Figure 6.6.

The approximate peak noise amplitudes measured for each hardware configuration are summarized in Table 6.1. Adding nine 1 μF ceramic capacitors significantly reduced both the current and voltage noise. Subsequently, the six additional 100 nF ceramic capacitors further reduced the current noise, although the input voltage increased but it may be due to imprecise measurements from the oscilloscope. x

Table 6.1: Approximate noise amplitudes measured for the three hardware configurations. I do not like how these values were measured. In reality the values are less than a quarter. They must be measured separately with a good trigger. I may redo it in V1.3.

Quantity	Original	+9 $\times$ 1 μF	+9 $\times$ 1 μF +6 $\times$ 100 nF	...+ schottky diode
I_{in}	1.8 A	640 mA	292 mA	140 mA
I_{out}	0.8 A	300 mA	220 mA	272 mA
V_{in}	2.0 V	350 mV	380 mV	490 mV
V_{out}	8.0 V	2.3 V	2.2 V	1.440 V

6.3.2 Efficiency and voltage and current ripples.

The efficiency of the MPPT charger is one of the most important parameters to be measured since it directly affects the power that is effectively extracted from the solar panel and delivered to the battery. As in simulations, the efficiency was measured for each point of the charging curve. All the measurements were taken with the multimeter and the results are presented in Table 6.2.

The measured efficiency is more than 96% for all test cases, which is a very good result. This efficiency takes into account all the losses in the converter, including the measurement system and control system. When compared with the power losses estimation made in section 5.2, the efficiency is

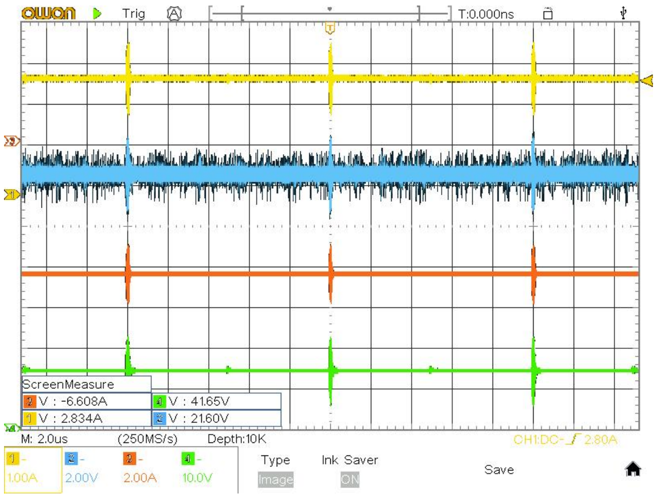

Figure 6.4: Measured waveforms before adding the additional ceramic capacitors.

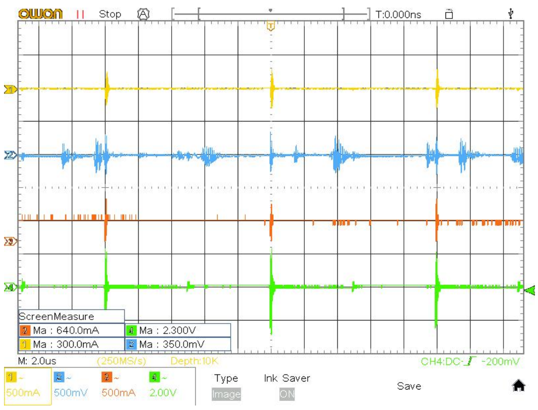

Figure 6.5: Measured waveforms after adding nine additional 1 μF ceramic capacitors.

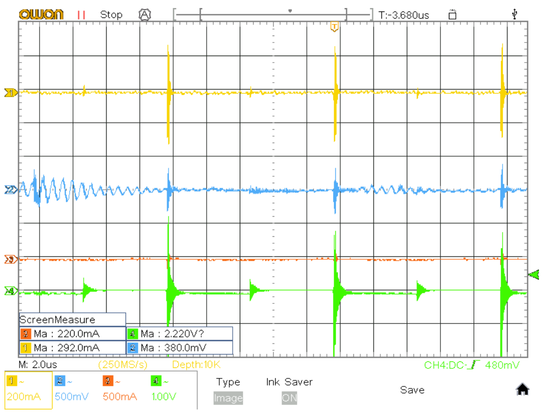

Figure 6.6: Measured waveforms after adding nine additional 1 μF ceramic capacitors.

Table 6.2: Efficiency measurements for the different test cases.

	Multimeter			- OC power	- OC+Transistors power
Case	Power in	Power out	Efficiency	Efficiency	Efficiency
Case 1	127.444	123.413	96.84%	97.23%	97.64%
Case 2	129.108	124.685	96.57%	96.96%	97.36%
Case 3	127.523	123.232	96.64%	97.02%	97.43%
Case 4	73.1277	70.818	96.84%	97.52%	98.24%
OC	0.5103	0	–	–	–
OC+Transistors	1.04279	0	–	–	–

around 2% lower than expected. This difference can be explained by the fact that the components used can have slightly different characteristics. Also, the losses caused by the parasitic elements of the PCB were not considered in the estimation.

The PCB was also tested in open circuit with and without driving the transistors. The results show the losses of the circuit without the boost converter. Around 0.5 w of losses are caused only by the measurement and control system. When the transistors are driven without any load, the loses of the converter are negligible since the current is too low. So the measured losses are approximately equal to the driving circuit + the measurement and control system. Based on these values, in table 6.2 it is also present a estimate of the efficiency without the measurement and control system and another without the driving circuit too. The efficiency is around 97% to 98% for all test cases, which is a very good result for a boost converter. From another perspective, it shows just how significant the losses are on the rest of the circuit.

6.3.3 Temperature rise

The temperature of each element was measured before each test and after 10 min. All test show similar result, a temperature rise of up to 10ºC. The inductor is the main heat element, reaching 43ºC, the capacitors also reach around 40ºC hover they are placed side by side with the inductor which heats the capacitors. As expected, the transistors do not overheat. The maximum temperature measured was 39.5ºC.

The results are close to what was expected. In Section 5.2, we concluded that the component with the highest losses is the inductor, followed by the capacitors and the transistors. In this section, it is expected that the temperature follows the same order since it is related to the losses. However, the measurements do not follow the expected results. This is explained by the fact that each of these components have different surface areas, and therefore there heat dissipation is not comparable.

We were also able to obtain thermal images of the circuit (Figure 6.7 and 6.8), which show the areas with the highest temperatures. Unfortunately, the image is of poor quality and shows a vertical shift between the thermal image and the normal image. Additionally, the device show some vertical lines that are a defect from the measuring device and not an actual reading. In any case, it appears that the inductor is the element that heats up the most, it is also seen a clear heat spread coming from it. The capacitors also show some heat dissipation but not from the same magnitude. On the back side, the only element heating is the transistors.

To confirm that the temperature does not rise to a concerning level, a test of 3 hours was performed with the converter operation at Case 3. In figure 6.10 it is represented the temperature evolution over the 3 hours for the inductor, capacitors, transistors and the ambient temperature. These results were acquired using the thermistors placed on the board which measure the ambient temperature and the transistors' temperature. The other measurements where acquired using additional external thermistors that can read by the board using the connector placed for thermistors.

The measured values were confirmed using an infrared thermometer which show a slightly different temperature, confirming the quality of the measurements.

Figure 6.7: Front side thermal image.

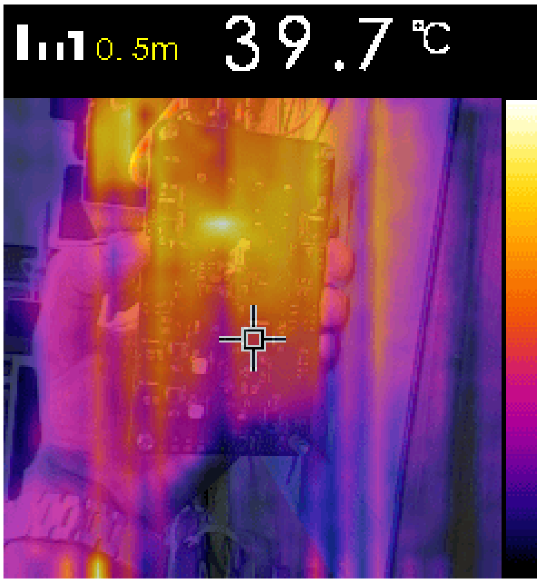

Figure 6.8: Back side Thermal image.

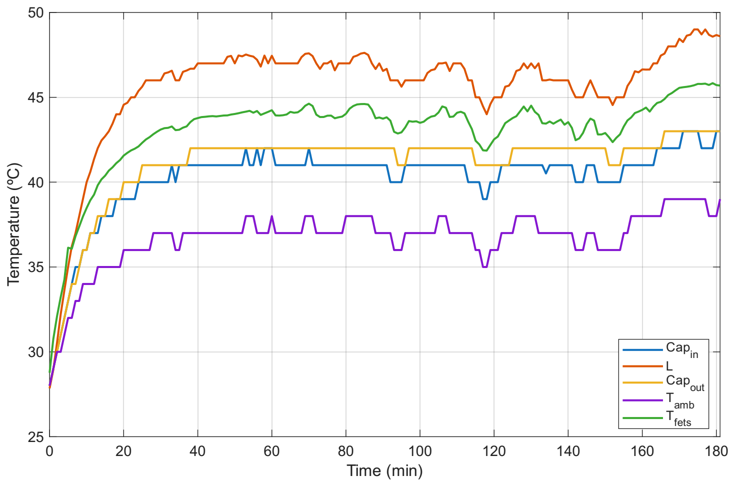

Figure 6.9: Temperature evolution during 3 hours.

From Figure 6.10, it can be concluded that after 40 min at maximum power, all the temperatures stop rising. Some oscillations were measured during the experiment due to the fact that ambient temperature also oscillated. As seen earlier the inductor is the element which heats the most, followed by the transistor and capacitors. After 3 hours, none of the elements reached a concerning temperature.

With this test it is concluded that there is no need for cooling. At least in the same conditions of the test. To confirm this statement, the board was placed inside a cardboard box since it is how TSB usually places boards in a vessel. If the components inside the box need cooling, an Fan or a water cooling place is added.

In Figure 6.10 is represented the described test. Since there is no air flow, the ambient temperature

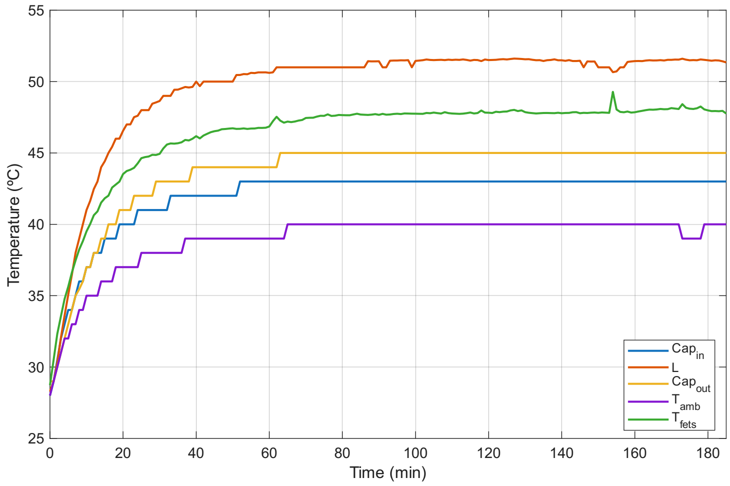

Figure 6.10: Temperature evolution during 3 hours in a close box.

reading is more constant and therefore the others too. As in the previous test, after 60 minutes the temperatures remained approximately constant and the maximum temperature was 51.6ºC. Since none of the temperatures show concerning values, it is safe to conclude that no additional cooling is needed

6.4 MPPT

In this section the charging sate machine is tested as well as the MPPT algorithm to ensure the functionality of the prototype.

In the following tests a laboratory setup was used, where a power supply (the SM 1500-CP-30) was used with an internal function which can simulate the behavior of a solar panel. This function was configured as a 35 cells solar panel at 25ºC and variable irradiance controlled by sending TCP messages.

The load was the battery from São Rafael 03 (SR03), which is described in section 2.3.

This setup ensures a controlled environment where the solar panel works at a specified temperature and irradiance parameters increasing the number of controlled variables. Since the environment conditions are set by the user, this setup can be easily replicated for future comparisons and therefore increasing the scientific value of the tests.

6.4.1 Efficiency

6.4.2 Tracking speed

For this test in specific, a python script was written which sends TCP messages (using LAN connection) to the power supply to change the irradiance periodically. First, the script sets the irradiance to

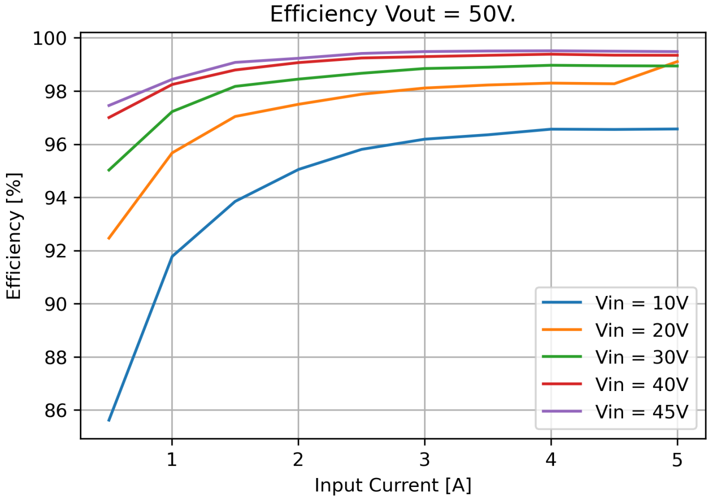

Figure 6.11: Converter efficiency over different input currents and voltages. Just a place holder image. No real data!!

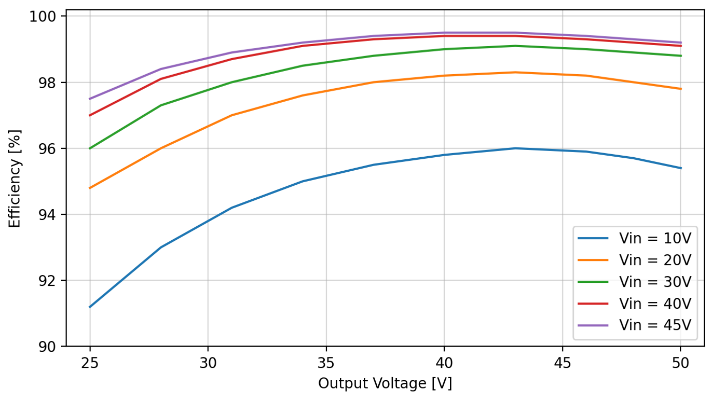

Figure 6.12: Converter efficiency over output voltage for different input voltages. Just a place holder image. No real data!!

200 $\mathrm{W/m^2}$. Subsequently, the irradiance is increased by 200 $\mathrm{W/m^2}$ with 2 seconds intervals until a final value of 1000 $\mathrm{W/m^2}$ is reached.

It is worth noting that the timing is not perfect and therefore the steps do not have perfect 2 seconds intervals. This is caused by a delay between the instant of which the power supply receives the message from the computer and actually changing the parameters.

The result represented in Figure 6.13 show an almost instantaneous response from the Perturb and Observe Algorithm. The algorithm is working at 100 Hz and with steps of 0.1% of duty cycle.

The result shows some noise, however they were acquired by the sensors present on the prototype and therefore the noise is from the sensors and not from the algorithm.

Overall, the algorithm showed a satisfactory fast response. In almost every scenario the algorithm act immediately except in step from 600 $\mathrm{W/m^2}$ to 800 $\mathrm{W/m^2}$, which is not a consistent and therefore

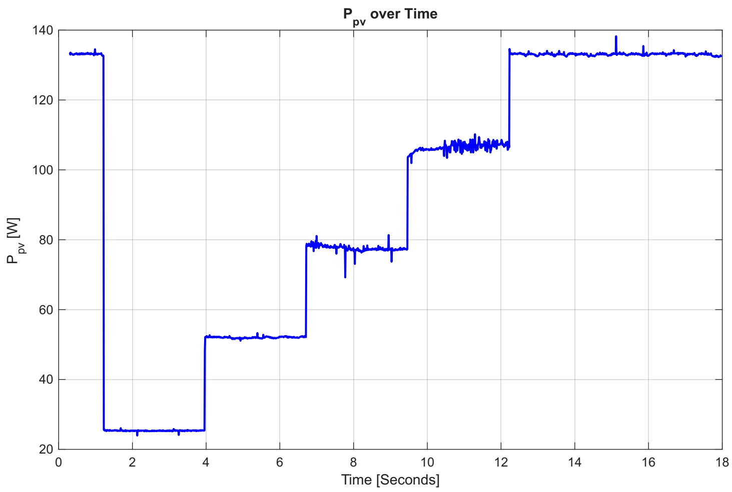

Figure 6.13: PV Power over time with steps of 200 $\mathrm{W/m^2}$, from 200 to 1000 $\mathrm{W/m^2}$.

there is no obvious explanation. Confirm this statement.

6.4.3 State machine

Show the MPPT in difrent states. No load-Sweep-CC-CV-no load-CV-CC

6.4.4 Battery charging

Do a full charge, validate CC-CV, battery temperature, MPPT temperature and robustness.

Como é que eu vou fazer isto!?! Vai demorar imenso tempo. Talvez faça com uma bateria da NAVICTUS ou do SP01. Tem menos capacidade e por isso demoram menos tempo a carregar.

6.5 Field test

Ou teste em terra com um painel solar e uma bateria. Ou um teste em em água. Ou os dois.

7

Conclusion

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last thing to write

7.1 Future Work

last thing to write

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